When Yield Curves Invert Together:

Currency Crash Risk*

Alfredo Effendy†

University of Washington

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Abstract

A yield-curve inversion is a well-documented leading indicator of domestic slowdown, but a single inversion does not reveal whether the expected slowdown is country-specific or global. This distinction, however, decisively predicts the asset market response to the impending slowdown. In a portfolio of currencies, such as the carry trade, currency-specific risk might be diversified, while systematic risk is affecting every risky currency. I show that a signal constructed from clustered inversions predicts crashes in high-interest and high-beta currencies, and the carry trade. This result plays a role in giving an early warning to avoid the well-known carry trade crashes. Further conditioning on the slope shape of global yield curves refines the signal to distinguish between inversions driven by slowing growth expectations versus inflation-fighting monetary policies. The latter comes with a higher interest rates environment, which provides a higher compensation for the carry trade crash risk.

JEL: E43, F31, G01, G11, G15.
Keywords: Inverted Yield Curve; Disaster Risk; Exchange Rate; Currency Carry Trade.

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†Email: `aeffendy@uw.edu`.

1 Introduction

The currency carry trade borrows in low-interest-rate currencies and invests in high-interest-rate ones. It has been profitable on average because high-interest-rate currencies do not depreciate enough to offset the interest rate differential (IRD), in violation of uncovered interest parity (UIP) (Hansen and Hodrick, 1980; Fama, 1984; Engel, 2016). The modern literature interprets this premium as compensation for depreciation risk and organizes currency returns into priced factors (Lustig and Verdelhan, 2007; Lustig et al., 2011; Menkhoff et al., 2012). The premium accumulates gradually, but much of it can disappear during currency crashes arising from unwinds of carry trade positions and funding spirals. These crashes create negative return skewness (Brunnermeier et al., 2008), the carry trade has exhibited across various financial market and monetary regimes in the past century (Accominotti et al., 2019). In disaster models, these losses are the realizations for which the interest rate differential compensates investors in advance (Farhi and Gabaix, 2016; Chernov et al., 2018). For a carry investor, the question that matters is whether an impending crash is predictable.

Answering that question requires an indicator of when currency downside risk is elevated. Government yield curves encode market expectations about economic activity in real time, and an inversion has long been understood as a recession signal (Harvey, 1988; Estrella and Hardouvelis, 1991; Campbell and Shiller, 1991; Estrella and Mishkin, 1998; Bauer and Mertens, 2018). I find that a signal constructed from cross-country clusters of fresh yield curve inversion predict currency crashes. Although the signal is only active in about a fifth of the sample, it predicts four of the give worst months in carry trade. The losses are concentrated in high-interest-rate and high-beta currencies. Figure 1 illustrates the relationship between the signal and carry trade crashes.

This finding suggests that the bond market leads the currency market, with yield curve inversions providing an early warning of currency losses. Two features of the signal help

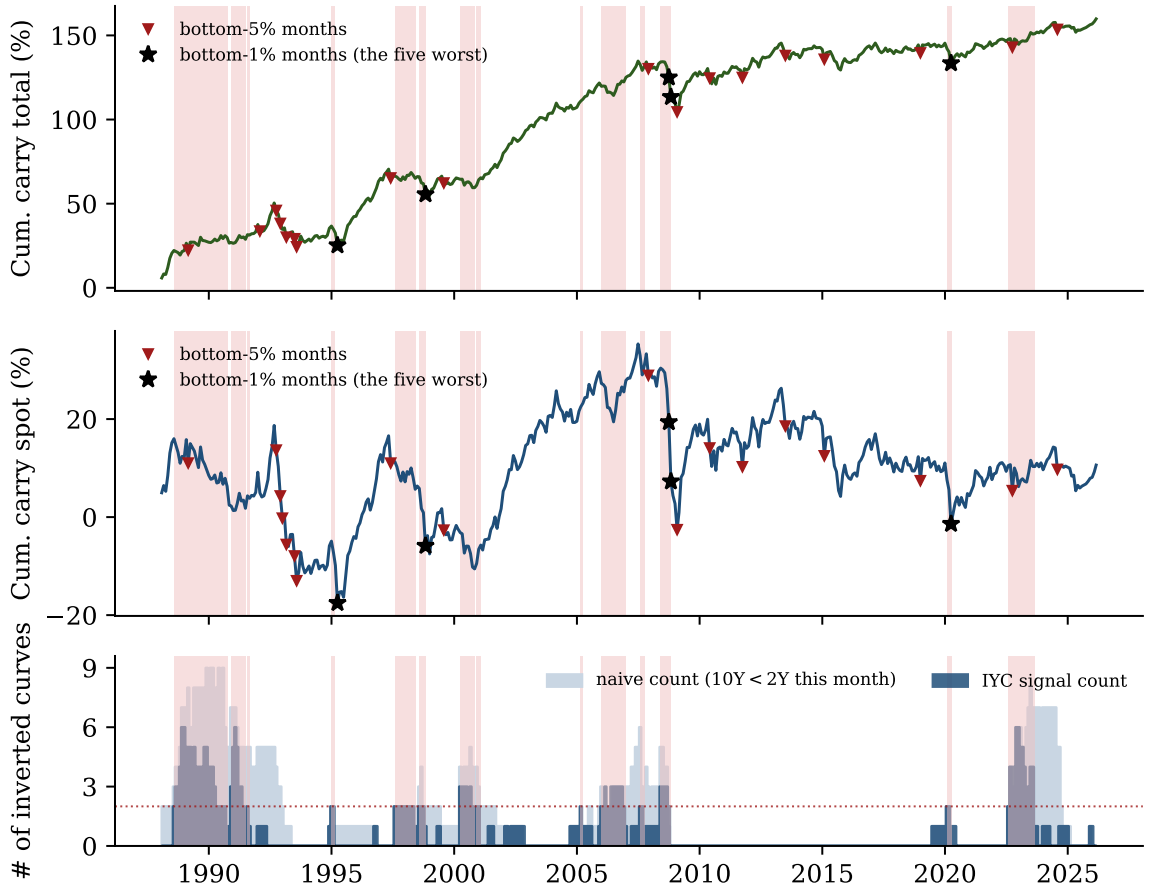


Figure 1: The carry trade and the IYC signal. Top to bottom, the carry portfolio's cumulative total return, its cumulative spot return, and two counts of inverted G10 curves. The *naive count* (light) counts curves with the 10-year yield below the 2-year this month, and the *IYC signal count* (dark) counts the inversions the signal counts, entered fresh and not yet confirmed re-steepened. The cluster threshold of two is dotted and IYC signal episodes are sh aded throughout. Triangles mark each return series' own bottom-5% months, the crash months, of which the two sets share 22 of their 23. Black stars mark its bottom-1% months, the five worst of the sample, four of which fall inside the traded state.

explain its predictive content. First, looking at clustered instead of individual inversions helps to distinguish a global risk from country-specific ones, which matters less to a diversified cross-country portfolio such as the carry trade. Consistent with this distinction, months with exactly one active inversion remain favorable for the carry trade.

Second, an inversion tends to persist, sometimes for years. If the currency market has adjusted to the outlook signaled by the inversion, it might lose its predictive power. The

simplest way to detect a cluster is to count how many curves are inverted in a given month, a measure that I call the *naive count*. This captures clustering, but treats fresh and stale inversions alike, potentially diluting the signal. Therefore, it makes sense to specifically look at clusters of fresh inversion.

There are several ways to account for freshness. This paper focuses on an approach that uses the information from the yield curve itself. A curve becomes active when it first inverts and remains active until its slope has steepened for two consecutive months, whether or not the slope has returned above zero. Steepening indicates that either a recovery is underway or the market has begun to price in the inversion, while the two-month confirmation follows the two-quarter recession conventions used in the United Kingdom, Canada, and Australia. Requiring the confirmation also handles curves whose slope hovers around zero, which might otherwise create a noisy alternating inversion state. An alternative approach is to penalize older clusters by weighting the naive count by the inverse of the cluster's age, which Appendix B.2 reports

I construct the *IYC signal*, from the G10 government yield curves, with both distinctions above built in. The baseline configuration for the yield curve slope is the ten-year yield minus the two-year yield, and Appendix B.1 repeats the construction with the ten-year minus the three-month yield. I use a simple threshold of two active inversions to separate a country-specific inversion from a cluster, since two is the smallest count that rules out a single-country event. The signal is on in 93 of the 458 sample months ($\sim$ 20%), across ten different episodes. Figure 1 shows how the freshness consideration separates the IYC signal from the naive count, which stays active for more months, and Section 2 gives the full construction of the signal.

Persistently high-interest-rate economies are often commodity exporters and small open economies that are exposed to the global cycle (Ready et al., 2017; Della Corte et al., 2016).[1]

[1]Complementary accounts based on country size (Hassan, 2013), position in the trade network (Richmond, 2019), and the static ranking of Hassan and Mano (2019) load many of the same currencies on world risk.

The carry trade inherits this exposure because it is long high-interest-rate, high-beta currencies and short low-interest-rate, low-beta currencies. When the signal is active, a portfolio that is long high-beta currencies and short low-beta currencies (beta HML) loses 9.24% annualized in spot terms (8.29% including IRD), against a gain of 3.57% (5.29% including IRD)when it is inactive. Similarly, the carry trade spot leg loses 8.99% annualized (3.67% including IRD) against a gain of 2.64% (6.19% including IRD). By contrast, a currency whose own curve is active while the signal is off does not depreciate on average, and months with exactly one active inversion remain favorable for the carry trade, returning 10.09% annualized.

The signal changes more than the average return. During these signal months, portfolio skewness becomes more negative and returns deteriorate throughout the distribution, with the largest changes in the left tail, accounting for much of the negative skewness of the carry trade. Composite leading indicators (CLI) also decline across the G10 economies after the signal months, including in the United States. Both the exchange rate and macroeconomic results show that the depreciation in high-interest-rate high-beta currencies and declining are due to the bond market pricing in a systematic weakening in economic activities, rather than one country domestic slowdown. In the language of the disaster-risk literature (Rietz, 1988; Barro, 2006; Wachter, 2013), I interpret the IYC signal as an observable indicator of elevated currency downside risk.

An inversion can occur in two different environments. First, market participants may expect rates to fall because of weakening economic activity (growth-scare); second, central banks may have or be in the process of raising rates to fight inflation, which market participants may read as transitory. A growth-scare inversion usually occurs in a low-interest-rate environment that generates a low compensation to potential depreciation in the carry trade. An inflation-fighting inversion occurs in a high interest rate environment, providing a large compensation for the carry trade. In the latter case, the IRD compensates the trader enough to offset the risk of a crash.

The shape of the yield curve helps to distinguish these cases. When the three-month yield is higher than the two-year yield and the two-year yield is higher than the ten-year yield, it represents an environment where the short-term rate is the highest and the market is expecting the interest rate to go down systematically in the medium and long term. I call this strictly downward-sloping configuration the *downcurve* and use the average G10 yield curve to measure it. In the months with this downcurve shape, the IRD compensates most of the depreciation that follows, while in other cases, it only compensates a small fraction of the depreciation. One can think of the signal as an early warning of an elevated crash risk, while the downcurve indicates whether the risk is adequately compensated.

A follow-up question is how to apply the constructed signal in real-time portfolio risk management. The natural response is to unwind the position when the signal is active to hedge the upcoming crashes. Doing so however, forgoes the IRD, which in months with downcurve configuration, compensates most of the losses. I call the risk of surrendering this compensation *carry-premium risk*, and the decision to go against the signal to stay invested the *carry overwrite*. Other options that investors and traders can do are hedging the tail via options Jurek (2014) or going anti-carry (Caballero and Doyle, 2012) during these signal months.

One potential concern is that the signal only repackages the global yield curve, the U.S. yield curve, or standard risk and volatility predictors such as the VIX, FX volatility, commodity index, and the NFCI. I therefore run a horse race regression against all those measures, including each country's inversion, the U.S. included, and the results' power survives or even strengthens in some cases.

The signal is also distinct from the curve-based predictors existing in the currency literature. The closest yield-curve studies use relative Nelson-Siegel factors, curvature, and other shape measures to predict bilateral exchange rates and currency returns Chen and Tsang

(2013); Dreher et al. (2020); Lloyd and Marin (2024); Lustig et al. (2019). [2] These measure the difference between two countries' curves and remove the component they share, which is what we are interested in detecting. A pairwise result against the dollar also does not map directly into a diversified currency portfolio.

Conditioning carry trade performance on observable states is not new. Clarida et al. (2009) and Christiansen et al. (2011) allow the carry trade's returns and risk exposures to vary in different volatility and liquidity regimes, while Barroso and Santa-Clara (2015) construct currency portfolios from observable covariates. Other related works condition on average interest rate levels (Lustig et al., 2014), inflation expectations (Andrews et al., 2024), and commodity, volatility, and liquidity predictors (Bakshi and Panayotov, 2013). The novel element is the conditioning variable, where a fresh cross-country cluster of yield-curve inversions is used to identify an elevated crash risk in the currency market, and I am not aware of any prior work that uses this specific approach.

The remainder of the paper proceeds as follows. Section 2 presents the measure of risk compensation, and the portfolio and signal construction. Section 3 reports the predictive evidence, the cross-sectional exposure mechanism, the change in the return distribution, and the portfolio hedging application. Section 4 compares the signal with other curve measures and standard risk indicators. Section 5 extends the study to emerging market currencies.

2 Empirical Framework and Signal Construction

2.1 Carry Crashes and Common Exposure

The portfolio construction follows the setup in Lustig et al. (2011), from a U.S. investor's perspective. Let $s_{i,t}$ be the log of the spot price of currency i against the dollar and $i_{i,t}$ be the annualized one-month money market rate. Under UIP, the expected depreciation

[2] Yield curves and currencies have also been priced jointly in affine models Backus et al. (2001); Ang and Chen (2010).

(appreciation) would offset the IRD collected (paid), so a long position in currency i funded in dollars would earn nothing in expectation.

$$\mathbb{E}_t[\Delta s_{i,t+1}] \;=\; -\tfrac{1}{12}\left(i_{i,t} - i_{US,t}\right), \tag{1}$$

The carry trade literature documents the violation of UIP, which results in the profitability of the carry trade strategy. Thus, the excess return of the carry trade can be written as follows:

$$r_{i,t+1} \;=\; \underbrace{\tfrac{1}{12}\left(i_{i,t} - i_{US,t}\right)}_{\text{carry leg}} + \underbrace{\Delta s_{i,t+1}}_{\text{spot leg}} \tag{2}$$

Each month, the currencies are ranked based on their interest rates $i_{i,t}$, and the *carry trade* goes long the k highest interest rate currencies and short the k lowest interest rate ones, equally weighted in each leg. So, the carry trade is essentially a high-minus-low (HML) interest rate portfolio of size $k \times k$:

$$r_{t+1}^{carry} \;=\; \frac{1}{k}\sum_{i\in\, top-k} r_{i,t+1} \;-\; \frac{1}{k}\sum_{i\in\, bottom-k} r_{i,t+1} \;=\; C_t + X_{t+1}^{carry} \tag{3}$$

where applying the portfolio weights to the two components of (2) gives the carry leg $C_t = \frac{1}{12k}\big(\sum_{\text{top-}k} i_{i,t} - \sum_{\text{bottom-}k} i_{i,t}\big)$ and the spot leg $X_{t+1}^{carry} = \frac{1}{k}\big(\sum_{\text{top-}k} \Delta s_{i,t+1} - \sum_{\text{bottom-}k} \Delta s_{i,t+1}\big)$. I use $k = 3$ as the baseline for the nine G10 currencies and for the sixteen-currency portfolio including both G10 and emerging market currencies in Section 5.2; the 1×1, 2×2, and 4×4 constructions are provided in Appendix B.1.

The *beta HML portfolio* replaces the sorting variable with the *currency beta*, the currency's exposure to the equity market in the sense of Lettau et al. (2014) and Kremens and Martin (2019). For each currency, $\hat{\beta}_{i,t}$ is the slope from the rolling regression of daily appreciation on the daily S&P 500 return over the 90 trading days ending with month t,

$$\Delta s_{i,d} \;=\; \alpha_i + \beta_i\, r_d^M + \varepsilon_{i,d}, \qquad d \in \mathcal{D}_t, \tag{4}$$

where $\mathcal{D}_t$ is that window and r^M is the equity market return. The portfolio applies the construction of (2) to the beta ranking,

$$r^{\beta}_{t+1} \;=\; \frac{1}{k} \sum_{i \in \text{ top-}k(\hat{\beta}_{i,t})} r_{i,t+1} \;-\; \frac{1}{k} \sum_{i \in \text{ bottom-}k(\hat{\beta}_{i,t})} r_{i,t+1} = C^{\beta}_{t} + X^{\beta}_{t+1} \tag{5}$$

Appendix B.5 repeats the exercise using a monthly 24-month rolling estimator and MSCI World instead of S&P 500 to calculate the beta.

The income leg C_t is known at time t and is strictly positive. So, the carry trade loses in a particular month if and only if $X^{carry}_{t+1} < -C_t$. Hence, every carry crash is an exchange-rate event in which the spot leg depreciates significantly more than the IRD obtained. In the sample, the income leg averages +3.91%/yr, with a minimum of +0.7%/yr in any given month; the spot leg averages +0.28%/yr, but it is negatively skewed, containing the left-tail events of the carry trade strategy. (Section 3.1).

The carry portfolio diversifies idiosyncratic risk by construction, since it involves investing in multiple currencies and has regular rebalancing. Suppose the spot return of currency i is the sum of a common and an idiosyncratic component,

$$\Delta s_{i,t+1} \;=\; -\,\beta_i\, g_{t+1} \;+\; e_{i,t+1}, \tag{6}$$

where g_{t+1} is a realized common shock to the global cycle, such as a growth scare, and β_i is currency i's exposure to it. The currency beta of (4) measures this exposure empirically, using the equity market return as the proxy for g. Under the portfolio weights of (2), the spot leg is the exposure gap between the legs times the common shock, $-(\bar{\beta}^L - \bar{\beta}^S)\, g_{t+1}$, plus the difference of the legs' average idiosyncratic shocks. The second piece has a variance of order σ^2_e/k when each leg holds k currencies, so diversification and the monthly rebalancing keep it small. Consistent with this, the significance of the signal, which is a function of the standard error of the returns, widens systematically as we increase the number of pairs,

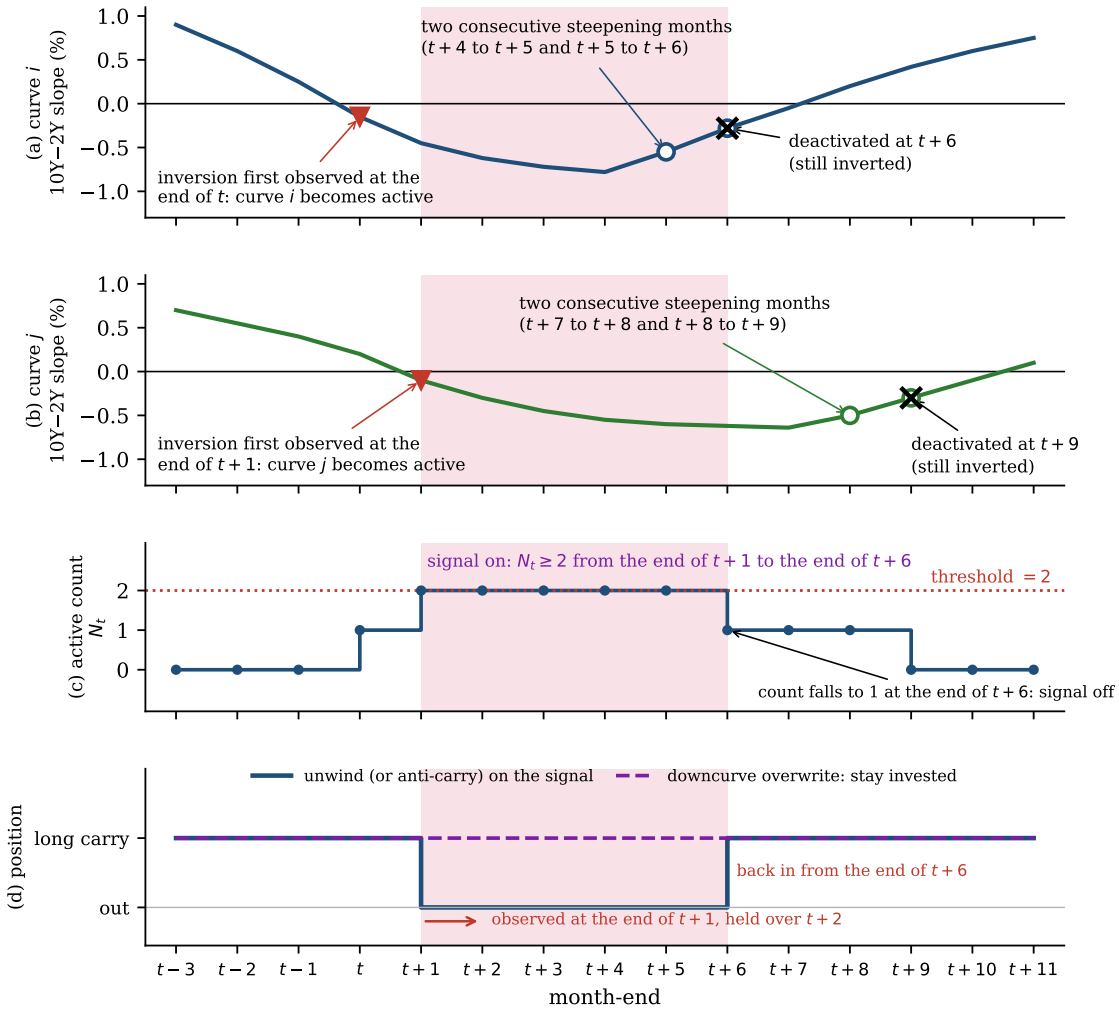


Figure 2: Signal timeline illustration. Panels (a) and (b) are one country's 10Y−2Y slope with the dated events of (8), where the inversion is first observed at the end of month t and the signal becomes active in that month (▼). Once its slope steepens in two consecutive months (circles), the signal becomes inactive (×) even though it is still inverted. Panel (c) is the active count N_t across the G10 curves against the threshold of two, and the shaded band is the signal state IYC_t. Panel (d) is the position, unwinding (or going anti-carry) for the shaded months and returning when the signal turns off, while the downcurve overwrite stays invested throughout. Figure 3 shows an example of a real episode.

with the 1x1 carry trade having the most idiosyncratic noise (Appendix B.1). Diversification bounds the size of the idiosyncratic piece but not its predictability. I therefore assume the

piece is unforecastable, $\mathbb{E}[e_{i,t+1} \mid \mathcal{F}_t] = 0$, where $\mathcal{F}_t$ contains the time-t yield curves and the portfolio sorts, and Section 4.1 tests this restriction's empirical implication for the paper's classifier. Only the common term carries a systematic loss, and the carry sort is exposed to it because the interest rate ranking tracks the exposure ranking (Section 3.2). Consequently, a warning signal for the carry trade must accordingly be informative about the common shock and not only about any single economy. Each country's yield curve prices its own economy's expected interest rate path so that a lone inversion can signal a country-specific risk. Requiring a cluster of inversions would eliminate the case of a country-specific risk. The data support the premise that these clusters of inversions predict real activity, as they lead the CLI.

2.2 Synchronized Yield Curve Inversions as a Signal of Systematic Risk

Let $y_{i,t}^{(n)}$ be country i's n-period yield. Under the expectations hypothesis with a slow-moving term premium τ_i, the slope between a longer-dated tenor L and a shorter-dated tenor S is

$$y_{i,t}^{(L)} - y_{i,t}^{(S)} = \underbrace{\frac{1}{n_L} \sum_{h<n_L} \mathbb{E}_t[i_{i,t+h}] - \frac{1}{n_S} \sum_{h<n_S} \mathbb{E}_t[i_{i,t+h}]}_{\text{expected short-rate path}} + \tau_i + \varepsilon_{i,t}, \tag{7}$$

where $i_{i,t+h}$ is the one-month interest rate, n_L and n_S are the two maturities in months, τ_i is the term premium differential between them, and $\varepsilon_{i,t}$ is the transitory deviation from the hypothesis. The slope is negative when the market expects the short-rate path beyond the shorter tenor to be lower than the one up to that tenor. This happens when activity is expected to weaken (Harvey, 1988; Estrella and Mishkin, 1998) or when the central bank policy rate is very high due to inflation fighting and the market is expecting it to fall. The alternative explanation, an inversion produced by the premium rather than by the expected path, is addressed in Appendix B.9.

The signal is built curve by curve. Let $v_{i,t} = y_{i,t}^{(L)} - y_{i,t}^{(S)}$ be a yield curve i's slope between the longer and shorter tenors of (7) at the end of month t, and $I_{i,t} = \mathbf{1}\{v_{i,t} < 0\}$ denotes its inversion. The baseline construction specifies L and S as the ten-year and two-year yields. Each curve carries an indicator $a_{i,t}$ for an *active* inversion and a counter $c_{i,t}$ for consecutive steepening months, $c_{i,t} = (c_{i,t-1} + 1)\,\mathbf{1}\{\Delta v_{i,t} > 0\}$, and

$$a_{i,t} = \begin{cases} 1 & \text{if } a_{i,t-1} = 0 \text{ and } I_{i,t} = 1,\ I_{i,t-1} = 0 \quad \text{(fresh inversion)}, \\ 0 & \text{if } a_{i,t-1} = 1 \text{ and } c_{i,t} \geq 2 \quad \text{(deactivated)}, \\ a_{i,t-1} & \text{otherwise}, \end{cases} \tag{8}$$

$$N_t = \sum_{i=1}^{9} a_{i,t}, \qquad \mathrm{IYC}_t = \mathbf{1}\{N_t \geq 2\}.$$

A yield curve inversion becomes active in the month its inversion is first observed. An active inversion is deactivated only after two consecutive months of steepening, whether or not the slope is going back above zero. A deactivated curve that stays inverted cannot trigger a new inversion signal until it has reverted and re-inverted. This information is observed at the end of month t; the count and the state IYC_t are updated after observing the information, and the updated position is held over to month $t + 1$. So, every return in which the paper regresses on the state is realized after the state is observed (no look-ahead bias). The baseline construction uses nine of the G10 yield curves, with the US curve and US dollar excluded to isolate the non-US signal. Table B.5 repeats the studies with the US curve included in the count and the dollar in the portfolio. It also provides an alternative signal construction using one and three consecutive steepening months instead of two. Appendix B.2 studies the naive count and an age-weighted inversion count as an alternative measure of freshness, which Figure 3 illustrates on one of the episodes. Figure 2 provides an illustration of the activation and deactivation of the signal.

If the IYC signal carries information about a common slowdown, that information should

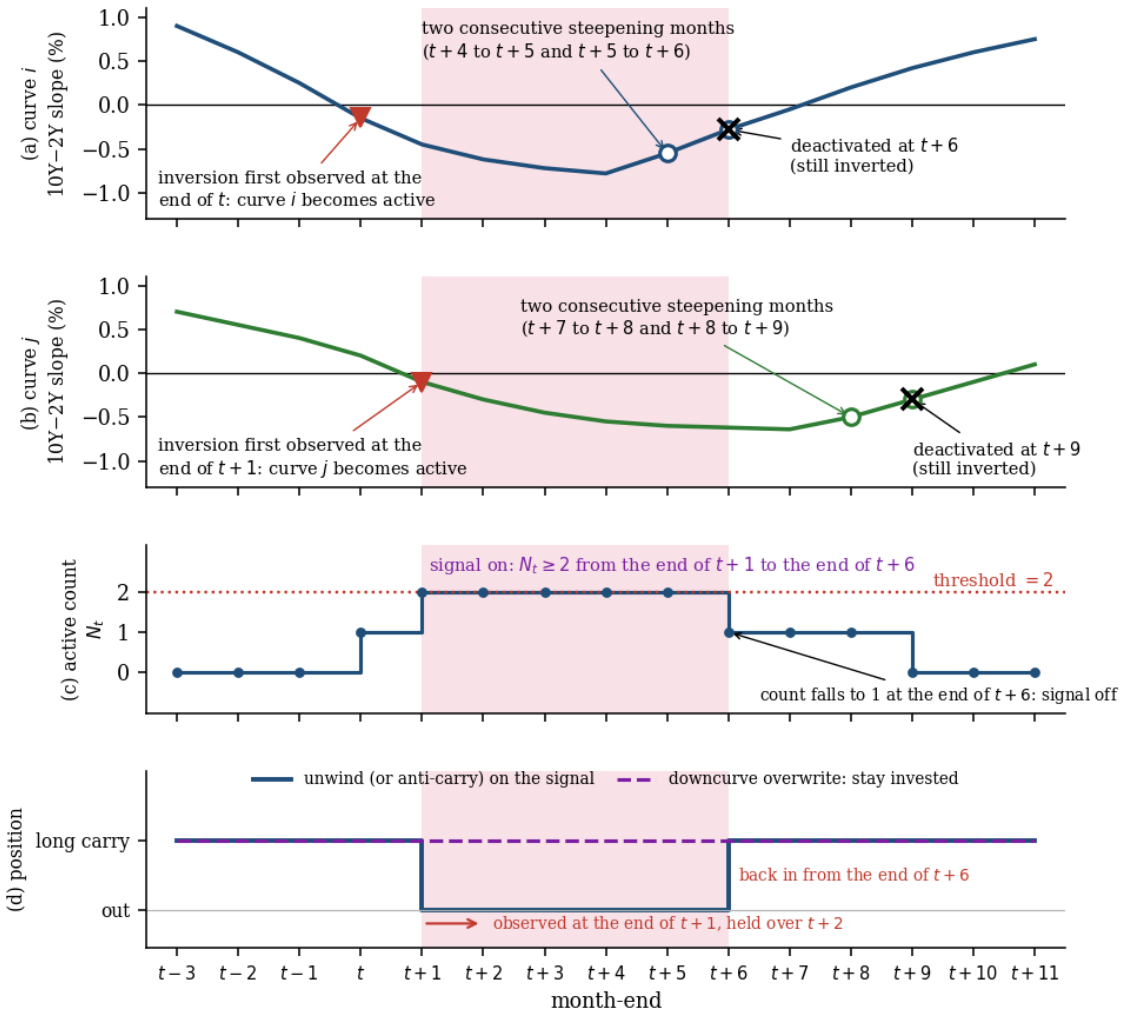


Figure 3: Sample of the IYC signal and naive count inversion comparison from 1998:01–2000:07. Panel (a) shows an example from Norwegian 10Y–2Y slope; both naive count (grey bar) and IYC (shaded) started the signal in August 1998 (▼), the slope steepened for two consecutive months in December 1998 (×) and IYC deactivated in January 1999 while naive count stays active until April 1999. Panel (b) shows the inversion counts comparison across the G10 yield curves for IYC and naive count. Panel (c) shows the cumulative returns of the carry trade (solid) and its spot leg (dashed), with the IYC and naive count shaded in red and blue, respectively. In the periods where the naive count is active and IYC is not, from November 1998 to April 1999, the carry trade actually had a positive return of 6.2%.

also have an implication in the indicator that leads activity and not only in currency returns. Table 1 shows how the IYC signal (or a cluster of inversions) affects the CLIs of the G10 economies, including the United States. [3] Table B.10 in Appendix B.3 repeats the exercise for the G10 currencies against the dollar.

Decomposing a country's own inversion into the ones overlapping with the IYC signal and those that are not, the latter ones do not predict their own CLI when pooled on 75 country-level months; similar results for currency, except that Norway has a significant opposite predictive result in this case. This shows that most of the predictive power comes with an active IYC signal, which is when a common systematic risk is building across different economies. Extending the exercise to predict the other G10 CLIs using one country's inversion yields the same results, further supporting the IYC predictive power on systematic slowdown. Appendix B.10 provides a local projection study to see what will happen in a longer horizon following the IYC signals, which leads to around one-half of a point lower within four months entering the IYC state. This result is consistent with evidence from Colacito et al. (2020) on currency premia compensating for business-cycle risk.

2.3 The Risk Compensation Ratio and Hedging

The natural application of the signal is to unwind the carry portfolio, or to reverse it into anti-carry, while the signal is active. An investor considering either of them would want to know the opportunity cost of this hedging, which is the carry income the position would stop collecting; more specifically, how it is compared to the crash risk. The benchmark for this is the Fama regression Fama (1984) of currency appreciation on the lagged IRD. It documents that in most of the months, a carry trade investor can collect the differential without being exposed to systematic depreciation, which leads to the carry trade being

[3] Canada, Japan, and Eurozone CLIs use the 10Y-3M yield spread as part of their calculation, excluding them from the study resulting in the same result, but with a lower significance (pooled coefficient of -0.089 and wild $p = 0.097$).

Table 1: Each economy's CLI after an own-signal inversion month and after an IYC signal month. ON counts the months where the country's own signal is active, and No Cl. counts the ones where the IYC signal is inactive. Each column reports the regression $A_{c,t+1} - A_{c,t} = \alpha + \gamma\,\text{State}_t + e_{t+1}$, where the states are each country's inversion signal (including and excluding the IYC signal) and the IYC signal, with the change in the OECD CLI as the left-hand side variable. I impose a minimum of four observations for the regression; results failing to meet that threshold would be denoted as –. The left column group reports the result for each country's CLI, while the right column group reports the result for the rest of the G10 CLI. The reported p-values are Newey and West (1987) (12-lag) and WCB ($B = 9,999$), while the pooled results add Driscoll and Kraay (1998) (12-lag) to take into account the country fixed effects. The sample runs from 1988:01–2026:02 (458 months).

			The Country's Own CLI			Rest of the G10, Average CLI		
	ON	Own Only	Own	Own, No IYC	IYC signal	Own	Own, No IYC	IYC signal
Australia	44	3	−0.045	—	−0.115	+0.026	—	−0.112
$NW_{12}[p]$			0.479	—	0.006	0.419	—	0.025
WCB$[p]$			0.642	—	0.096	0.552	—	0.032
Canada	41	10	−0.185	−0.039	−0.189	−0.100	−0.033	−0.104
$NW_{12}[p]$			0.000	0.478	0.000	0.086	0.264	0.032
WCB$[p]$			0.011	0.561	0.019	0.031	0.162	0.047
Switzerland	12	0	+0.063	—	−0.082	−0.010	—	−0.116
$NW_{12}[p]$			0.145	—	0.293	0.655	—	0.010
WCB$[p]$			0.751	—	0.264	0.739	—	0.033
Eurozone	28	0	−0.025	—	−0.100	−0.077	—	−0.114
$NW_{12}[p]$			0.616	—	0.140	0.111	—	0.013
WCB$[p]$			0.618	—	0.133	0.686	—	0.033
United Kingdom	68	21	−0.026	−0.065	−0.055	−0.054	−0.012	−0.119
$NW_{12}[p]$			0.764	0.194	0.471	0.230	0.752	0.009
WCB$[p]$			0.778	0.211	0.484	0.348	0.766	0.018
Japan	22	2	−0.092	—	−0.131	−0.088	—	−0.111
$NW_{12}[p]$			0.141	—	0.000	0.175	—	0.029
WCB$[p]$			0.244	—	0.004	0.698	—	0.060
Norway	70	27	−0.127	+0.100	−0.142	−0.120	−0.072	−0.109
$NW_{12}[p]$			0.076	0.629	0.149	0.067	0.309	0.016
WCB$[p]$			0.368	0.819	0.191	0.027	0.403	0.049
New Zealand	54	12	−0.115	−0.073	−0.068	−0.027	+0.050	−0.118
$NW_{12}[p]$			0.238	0.262	0.346	0.401	0.355	0.017
WCB$[p]$			0.368	0.424	0.396	0.391	0.611	0.020
Sweden	24	0	−0.115	—	−0.105	−0.114	—	−0.113
$NW_{12}[p]$			0.234	—	0.052	0.184	—	0.020
WCB$[p]$			0.284	—	0.069	0.101	—	0.029
United States	42	9	−0.023	+0.016	−0.130	+0.007	−0.025	−0.111
$NW_{12}[p]$			0.547	0.731	0.004	0.834	0.420	0.023
WCB$[p]$			0.461	0.782	0.022	0.915	0.546	0.038
Pooled G10			−0.085	−0.009	−0.110	−0.064	−0.025	−0.113
WCB$[p]$			0.019	0.957	0.039	0.005	0.473	0.036
DK$[p]$			0.022	0.906	0.025	0.031	0.375	0.018

profitable on average. So, hedging in normal times is expensive on average, and the question is whether the IYC signal can provide the periods in which hedging is justified.

Conditioning (2) on the IYC signal, we can rewrite the expected return of the carry trade as the income leg plus the expected spot leg. Assuming that there is an elevated crash risk for the spot leg conditioning on the signal, we can proxy it as a crash of size J arriving with

probability λ_t, so that $\mathbb{E}_t[X_{t+1}^{carry}|\text{IYC}] = -\lambda_t J$ and

$$\mathbb{E}_t\left[r_{t+1}^{carry} \,\middle|\, \text{IYC}\right] \;=\; C_t - \lambda_t J \;\lessgtr\; 0 \quad\Longleftrightarrow\quad \kappa_t \;\equiv\; \frac{C_t}{\lambda_t J} \;\lessgtr\; 1, \tag{9}$$

so, unwinding the portfolio or going for an anti-carry position raises the expected return if and only if $\kappa_t < 1$.

Neither λ_t nor J is observable at time t; the investor estimates them from historical data of previous signal months during the time of the observation. In Table 2, the IRD is higher in months where the signal is active, earning 5.32%/yr against +3.55%/yr when the signal is inactive. The compensation increases as the risk increases, but it is not enough to compensate for the spot depreciation during these periods (11.63% annualized). Therefore, it is justified if the investor decides to unwind the portfolio. Unwinding the portfolio, however, forgoes the IRD income and the spot return $C_t + X_{t+1}^{carry}$. There is an opportunity cost of doing it, especially when the carry income is strictly positive; the longer the unwind is, the more income the investor passes on, so the opportunity cost of unwinding increases as well. Doing an anti-carry on the other hand, only pays off if the spot leg crashes, but it costs more if it does not, since the investor is technically paying for the carry income instead of earning it. In both cases, the hedge forgoes the compensation for the risk it removes, and I call this the *carry-premium risk*, which is largest when the IRD is the widest. An investor can therefore decide, from the observed C_t and an estimate of the crash risk, whether to unwind the position or to stay invested and *overwrite* the signal. The decision would depend on the investor's risk profile. One way to differentiate the risk profiles in this case is to use κ from (9) above; a risk-averse investor would require a higher κ and vice versa.

2.4 The Downcurve Regime

The second case of Section 2.2 is when the central banks are fighting against inflation, leading to an unusually high short-term interest rate. However, this policy tends to be transitory,

meaning the market would expect the interest rate to be lower in the medium and longer term. When we look at the yield curves, this can be observed when the three-month yield, a proxy for a short-term interest rate, is higher than the two-year yield, a proxy for a medium-term interest rate. Additionally, the two-year yield is also higher than the ten-year yield, a proxy for a long-term interest rate. The paper calls this configuration *downcurve* and uses the average of the G10 curve to capture the cross-sectional policymaking:

$$\bar{y}_t^{(n)} = \frac{1}{9}\sum_{i=1}^{9} y_{i,t}^{(n)}, \qquad n \in \{3M,\, 2Y,\, 10Y\}, \tag{10}$$

so that the regime is

$$D_t = \mathbf{1}\{\bar{y}_t^{3M} > \bar{y}_t^{2Y} > \bar{y}_t^{10Y}\}, \tag{11}$$

The regime is captured in 52 of the 458 sample months, where 37 of them are inside the IYC signal. The split $\text{IYC}_t \times D_t$ is the object Section 3.5 evaluates to differentiate signal regimes. When the front end of the curve is not the highest point, the curve does not display the configuration associated with normalization from an inflation-fighting policy rate. I treat these remaining signal months as the growth-scare inversion. Figure 4 shows when the regime occurs. The downcurve months form two blocks, 1988–91 and 2023–24 (plus a single month in 2007), two well-documented high inflation environments. [4]

Figure 5 shows how the average yield curve's shape changes across the states, and Table 2 documents the bond yields, the money-market rates, inflation, and the carry trade's compensation in each of them. The downcurve months are the high-rate, high-inflation months of the sample whether or not the signal is on. Three-month yields are near 9.4% in the downcurve signal months and near 5% in the downcurve no-signal months (mostly the 2023–24 block), against about 4% in the other signal months, and the carry trade's IRD is at its largest there (+6.4 against +4.64%/yr within the signal). Panel B reports the inference

[4]The regime describes the environment in which an inversion arrives; the paper does not claim to identify what caused it.

Table 2: Stylized facts by state. Panel A reports the number of periods in months, the average three-month, two-year, and ten-year bond yields across the G10, the average money market rate across the G10, the money-market rate for the US, average year-to-year CPI inflation across the G10 (using the reading from two months earlier, a conservative approach for publication delay to make sure the data is observable at time t), the carry trade's IRD and spot return, and the carry trade's total return (all annualized, %/yr). Split by full sample, by IYC signal state, and by the downcurve regime. Panel B reports the gaps between the baseline IYC signal and with the downcurve filter applied to it. The reported p-values are Newey and West (1987) (12-lag) and WCB ($B = 9,999$), with the sample running from 1988:01–2026:02 (458 months).

		Rates and Inflation (%, Lagged)						Carry Trade (%/yr)		
State	n	3M	2Y	10Y	G10 Rate	U.S. Rate	CPI YoY	IRD	Spot	Total
Panel A: Mean by State										
Full sample	458	3.53	3.65	4.28	3.65	3.22	2.19	+3.91	+0.28	+4.19
non-downcurve months	406	2.94	3.13	3.90	3.06	2.71	1.86	+3.69	+0.41	+4.10
downcurve months	52	8.14	7.67	7.31	8.33	7.19	4.75	+5.62	−0.76	+4.86
No signal	365	2.84	3.00	3.80	2.96	2.50	1.84	+3.55	+2.64	+6.19
IYC signal	93	6.25	6.19	6.19	6.39	6.06	3.55	+5.32	−8.99	−3.67
non-downcurve months	56	4.18	4.39	4.67	4.25	4.92	2.57	+4.64	−14.95	−10.31
downcurve months	37	9.38	8.92	8.48	9.62	7.79	5.05	+6.35	+0.02	+6.37
Panel B: Gaps										
Downcurve − other, full sample		+5.21	+4.53	+3.41	+5.28	+4.48	+2.88	+1.93	−1.17	+0.76
$NW_{12}[p]$		0.001	0.004	0.022	0.001	0.000	0.000	0.042	0.657	0.756
WCB$[p]$		0.327	0.425	0.672	0.339	0.188	0.033	0.524	0.846	0.716
Signal − no signal		+3.41	+3.19	+2.39	+3.43	+3.57	+1.71	+1.77	−11.63	−9.86
$NW_{12}[p]$		0.002	0.002	0.010	0.003	0.000	0.002	0.004	0.003	0.013
WCB$[p]$		0.146	0.156	0.255	0.156	0.021	0.126	0.178	0.001	0.007
Downcurve − other, within signal		+5.20	+4.53	+3.81	+5.36	+2.86	+2.48	+1.71	+14.96	+16.68
$NW_{12}[p]$		0.000	0.000	0.000	0.000	0.000	0.000	0.040	0.011	0.005
WCB$[p]$		0.007	0.040	0.150	0.007	0.011	0.002	0.527	0.063	0.053

on the gaps. Within the signal, the downcurve gaps in money-market rates and inflation are statistically significant at the 5%. Over the full sample, only the inflation gap is significant, which might be due to the clustering nature of the regime. The differential gap is positive in every split but not significant. Of the signal-versus-no-signal gaps in rates and inflation, only the U.S. money market rate is significant (5%), with the inflation gap short of the 10% level ($p = 0.121$).

This compensation hypothesis links both terms of κ_t to the curve. The numerator is the IRD, which Table 2 shows at its highest in the downcurve configuration. The denominator is lower there too, because the crash in (6) is a growth scare and an inflation-fighting inversion

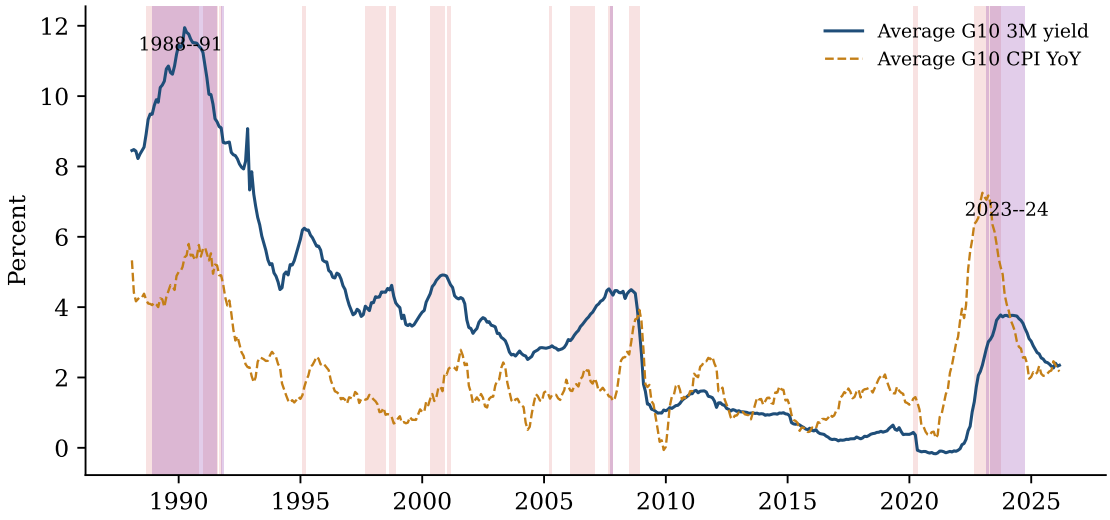


Figure 4: Downcurve, yields, and inflation. Average G10 three-month yield (solid) and CPI inflation (dashed); purple bands mark downcurve months (D_t of (11) on the average curve), pink bands the IYC signal episodes.

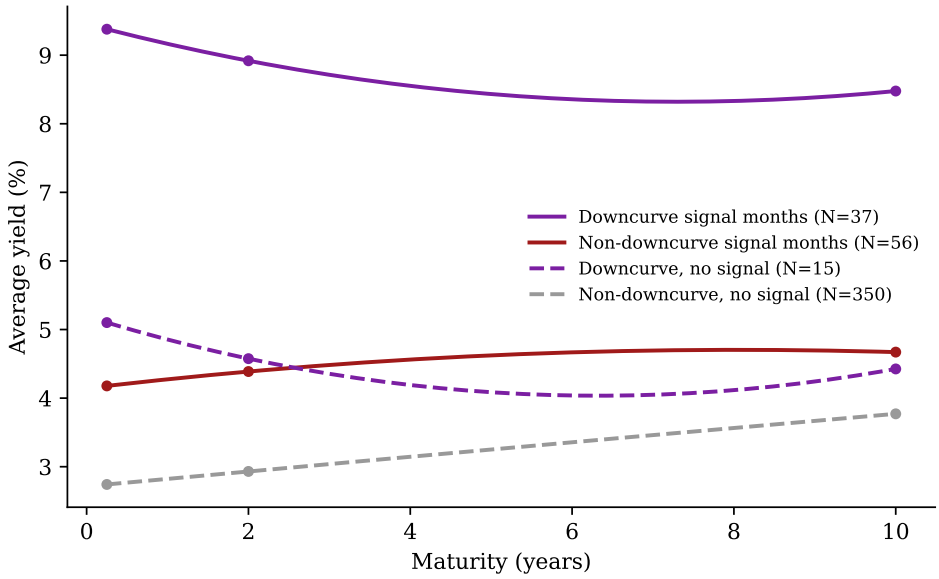


Figure 5: Average G10 yield curves by state and regime. The solid lines are downcurve and non-downcurve signal months; while the dashed lines are downcurve and non-downcurve no-signal months. Points are the average 3-month, 2-year, and 10-year yields.

carries less of one. In other words, the currency depreciation is compensated by the high IRD it offers, so these inflation-fighting inversion regimes follow the channel from Clarida and Waldman (2008). Bearing the signal's risk should therefore be *paid* in the downcurve

regime and *unpaid* in the rest, and Section 3.5 reports that the data follows this pattern. The overwrite strategy in 2.4 is exactly this, staying invested through those months instead of unwinding.

2.5 Data

The main sample uses monthly data and runs from January 1988 to February 2026. The baseline universe is the nine G10 currencies (AUD, NZD, CAD, NOK, SEK, GBP, EUR, CHF, and JPY), and I use the DEM as a proxy for the EUR before 1999. For each currency, I use its exchange rate against the dollar, its one-month money-market rate, and its 3M, 2Y, and 10Y government bond yields. The currency betas are estimated on daily (monthly) exchange rate and S&P 500 (MSCI World) returns, both of which are available 90 trading days (24 months) before the start of the sample.

For the EM extension, I add the seven most liquid floating currencies, BRL, CLP, IDR, INR, MXN, KRW, and ZAR. Due to data availability, their sample is shorter. The last money-market rate to start is the Chilean Peso, which started in January 1996. So, the EM panel begins in February 1996, while the alternative signal built on the EM curves themselves begins in June 2003, the first month when the EM curves are available.

Among the other variables, the NFCI, the CLI, and CPI inflation are available from January 1988, while the VIX begins in October 1990. The CLI ends in November 2019 for New Zealand and in November 2022 for Switzerland, Norway, and Sweden, and I extend each with that country's business or consumer confidence index, which affects the CLI result in some of the IYC signal months. All data are obtained from Bloomberg, Finaeon, central bank datasets, and the Chicago Fed. Appendix A sets out the economic regimes the sample spans.

3 IYC Signal Mechanism and Implementation

A synchronized yield curve inversion indicates an expected slowdown in different economies. We can think of it as a systematic risk building in the currency market; therefore, currencies that are exposed more to this risk (measured by the currency beta) are expected to depreciate more. The carry trade sorts the currency based on the IRD, which has a high correlation with the currency beta, which the beta HML portfolio sorts with. As a result, these portfolios are exposed to the systematic risk signaled by the IYC. This section shows this mechanism on the currency level through the IRD and currency beta ordering for the IYC signal (Section 3.2), then shows it in a portfolio setting (Section 3.1). Additionally, it also provides Fama regression conditioning on the IYC signal (Section 3.3), portfolio distribution conditioning on the IYC signal (Section 3.4), and the trading application of the IYC signal on the carry trade (Section 3.5).

3.1 Portfolio Predictive Regressions

The objective of the IYC signal is to predict the return in the next period after observing that the signal is active. The information required to construct the signals (IYC_t, D_t) is observable at the end of time t to predict the realized return at time $t+1$,

$$\text{Return}_{t+1} = \alpha + \gamma \text{State}_t + e_{t+1}, \tag{12}$$

where State_t consists of $[\text{IYC}_t, \text{IYC}_t \times D_t, \text{IYC}_t \times (1 - D_t)]$ to study how the signals perform when it is the original IYC construction, and when filtered with and without the downcurve regimes. I perform this predictive regression in Table 3 for four different types of portfolio settings: the carry trade's spot leg X_{t+1}^{carry} and total return r_{t+1}^{carry} from (3) and the beta HML's spot leg X_{t+1}^{β} and total return r_{t+1}^{β} from (5). The purpose of this distinction is to see how the signals affect the spot leg only compared to the total portfolio returns when

Table 3: Portfolio returns in periods where the IYC signal is active and inactive. The table reports the number of months, annualized return, Sharpe ratio, skewness, and the 5% expected shortfall (the mean of the worst 5% of the sample) for each state in each portfolio. Additionally, the predictive regression coefficient and intercept of (12) with the Newey and West (1987) (12-lag) and WCB ($B = 9,999$) p-values are also reported for the different configurations of the signals. The *ex bottom decile* rows provide a robustness check when removing the bottom decile returns (left tail) of the sample to see if the results still hold. Section 3.4 discusses the distributional shift in more depth. The sample is from 1988:01–2026:02 (458 months).

Portfolio	State	n	Mean (%/yr)	Sharpe	Skew	$ES_{5\%}$	α	γ (Gap)	$NW_{12}[p]$	WCB[p]
Carry, spot	All months	458	+0.28	0.04	-0.74	-5.68				
	No signal	365	+2.64	0.35	-0.52	-5.06				
	IYC signal	93	−8.99	-1.01	-1.05	-7.59	+2.64	−11.63	0.003	0.001
	non-downcurve	56	−14.95	-1.52	-1.10	-8.63	+2.40	−17.35	0.002	0.002
	downcurve	37	+0.02	0.00	0.27	-3.71	+0.30	−0.29	0.918	0.905
	ex bottom decile	80	−0.33	-0.06	0.76	-2.50	+8.05	−8.38	0.001	0.010
Carry, total	All months	458	+4.19	0.53	-0.68	-5.30				
	No signal	365	+6.19	0.83	-0.45	-4.69				
	IYC signal	93	−3.67	-0.41	-1.04	-7.24	+6.19	−9.86	0.013	0.007
	non-downcurve	56	−10.31	-1.05	-1.11	-8.27	+6.21	−16.51	0.003	0.004
	downcurve	37	+6.37	0.96	0.41	-3.00	+4.00	+2.37	0.351	0.396
	ex bottom decile	80	+5.09	0.85	0.81	-2.04	+11.53	−6.44	0.010	0.024
Beta HML, spot	All months	458	+0.97	0.11	-0.32	-5.78				
	No signal	365	+3.57	0.43	0.04	-4.85				
	IYC signal	93	−9.24	-0.95	-0.98	-7.97	+3.57	−12.82	0.001	0.006
	non-downcurve	56	−14.77	-1.62	-1.66	-8.60	+3.16	−17.93	0.000	0.001
	downcurve	37	−0.88	-0.09	-0.58	-6.83	+1.13	−2.01	0.740	0.858
	ex bottom decile	80	−2.96	-0.38	-0.21	-5.68	+6.58	−9.53	0.017	0.024
Beta HML, total	All months	458	+2.54	0.29	-0.27	-5.62				
	No signal	365	+5.29	0.63	0.07	-4.67				
	IYC signal	93	−8.29	-0.85	-0.92	-7.86	+5.29	−13.58	0.000	0.001
	non-downcurve	56	−13.29	-1.47	-1.60	-8.40	+4.74	−18.03	0.000	0.001
	downcurve	37	−0.71	-0.07	-0.50	-7.05	+2.82	−3.53	0.567	0.864
	ex bottom decile	80	−2.19	-0.28	-0.21	-5.75	+8.32	−10.51	0.008	0.006

taking into account the IRD. Additionally, Table 4 provides results of the same exercise at the currency level against the dollar to show how the signals relate to IRD and currency beta.

There are several properties of the data and signals that we need to consider when choosing the statistical methodology. First, monthly return errors are serially correlated and heteroskedastic, a common issue when dealing with financial time series data. Second, the signals are persistent; following the property of an inversion, once a yield curve is inverted, it tends to be inverted for several months. Third, inversions occur relatively infrequently, so

we are dealing with a rare event study. As a result, the constructed signal creates a block of signals across different time periods; we may have many signal months, but the effective number of periods is significantly smaller than that. The baseline IYC signal is active in 93 months, but the effective number of independent episodes is only between 5 to 14 (depending on the definition of an episode, whether it is the number of clusters, imposing a minimum gap between episodes, or based on the business cycle).

I provide two complementary p-values to address these concerns. The first one is using the wild cluster bootstrap (WCB) with Rademacher weights and imposing the null hypothesis (Cameron et al., 2008). This method holds the signal fixed and applies the same random weight, either (+1) or (-1), to the restricted residuals within each cluster. All the months belonging to the same episode receive the same weight, and it allows the errors to be correlated within the clusters, while assuming independence across clusters. I use replication $B = 9,999$ times following Davidson and MacKinnon (2000) for precision purposes. The second is the Newey and West (1987) p-value with 12-month lags, in order to capture the persistency of the signal, which accounts for the heteroskedasticity and serial correlation through the selected window.

The return for the carry trade portfolio mostly comes from the IRD or carry leg, yielding an annualized return of +3.91% compared to the spot leg of +0.28%. For the beta HML portfolio, on the other hand, the carry leg yields an annualized return of +1.57% and the spot leg of +0.97%. When the signal is not active, the carry trade and beta HML portfolio perform extremely well, with an annualized portfolio return of +6.19% and +5.29%, with +2.64% and +3.57% coming from the spot leg. So, the gain of the portfolios during an inactive IYC signal comes not only from the IRD, but also from the exchange rate appreciation of the high-interest-rate and high-beta currencies. On the other hand, when the IYC signal is active, the spot leg crashes, with the carry trade and beta HML's spot leg yielding annualized returns of −8.99% and −9.24%, respectively; Section 3.3 discusses this phenomenon in relation to the well-known Forward Premium Puzzle. Interestingly, the IRD from the carry trade portfolio

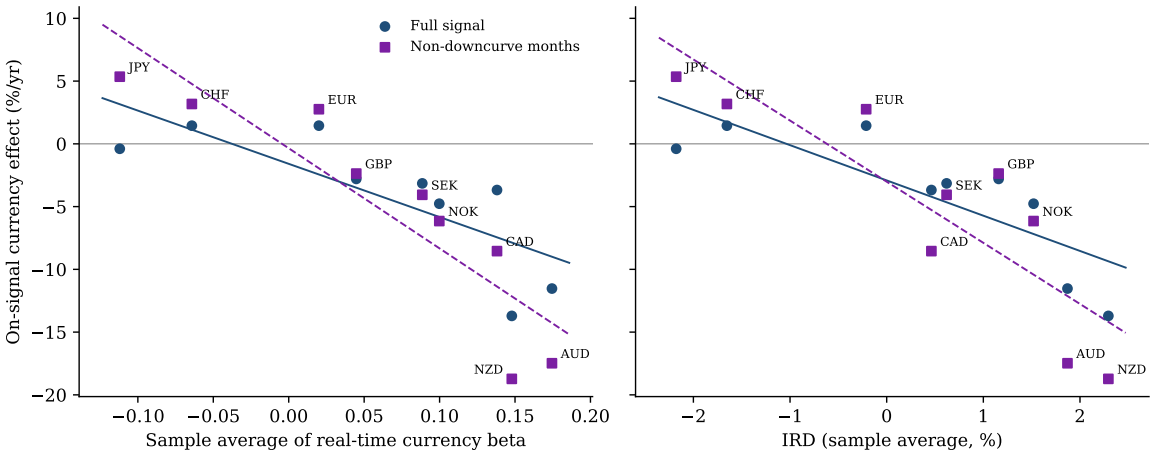


Figure 6: On-signal spot effect against the currency beta (left) and the average IRD (right), one point per currency, on a shared vertical scale. Circles and solid fits are for the full signal (slopes -42 and -2.8, $R^2 = 0.61$ and 0.66), squares and dashed fits for the non-downcurve months (slopes -80 and -4.9, $R^2 = 0.81$ and 0.75).

attempts to compensate for the higher crash risk during those periods, yielding an annualized return of +5.32%, higher than the full sample and no signal periods, but this is not the case for the beta HML portfolio. The predictive regression of the IYC signal on these portfolio returns is statistically significant at the 5% level for all constructions and methods.

When taking into account of the downcurve regimes within the IYC signal, in an attempt to differentiate growth scare and inflation-fighting inversions, the distinction is very apparent. The main driver of the crash is the growth-scare inversions, with the spot leg of the carry trade and beta HML portfolios yielding annualized returns of -14.95% and -14.77%, respectively. Meanwhile, within the downcurve regime, the spot return is basically flat, with the carry trade portfolio enjoying an IRD of +6.35% annualized, the largest income compared to the other periods, mostly due to the high inflation environment. However, there is a divergence in the beta HML portfolio, where the high-beta currencies are not yielding high interest rates during these periods, something to be explored in future studies.

Table 4: IRD and beta mechanism for the signals at the currency level. The first two columns after the currency column are the sample-average money-market IRD vs. the U.S. (%) and the sample-average currency beta (the daily 90-day rolling estimator of (4)). The next columns show the predictive regression for the signals at the currency level against the dollar: $r_{i,t+1} = \alpha_i + \gamma_i \, \text{State}_t + e_{i,t+1}$. The study is conducted for the spot only ($\Delta s_{i,t+1}$), and with IRD taken into account. The unit for the coefficients of the regression is annualized return (%). The number of months for each IYC, IYC without downcurve, and IYC with downcurve are 93, 56, and 37 months, respectively. The reported p-values are Newey and West (1987) (12-lag) and WCB ($B = 9,999$), with the sample running from 1988:01–2026:02 (458 months).

			Spot			Total		
Currency	IRD	β	IYC	Non-downcurve	Downcurve	IYC	Non-downcurve	Downcurve
AUD	+1.87	+0.17	−11.53	−17.48	+0.14	−10.68	−18.72	+3.78
$NW_{12}[p]$			0.056	0.045	0.980	0.090	0.028	0.519
WCB[p]			0.046	0.097	0.965	0.142	0.077	0.500
NZD	+2.29	+0.15	−13.71	−18.73	−2.81	−12.93	−18.89	−0.87
$NW_{12}[p]$			0.006	0.017	0.535	0.012	0.013	0.858
WCB[p]			0.007	0.068	0.423	0.016	0.062	0.826
CAD	+0.46	+0.14	−3.68	−8.55	+4.33	−3.20	−9.44	+6.68
$NW_{12}[p]$			0.260	0.057	0.108	0.358	0.031	0.017
WCB[p]			0.314	0.041	0.158	0.434	0.025	0.141
NOK	+1.52	+0.10	−4.77	−6.15	−1.50	−5.36	−7.60	−0.70
$NW_{12}[p]$			0.395	0.409	0.837	0.331	0.285	0.924
WCB[p]			0.443	0.488	0.861	0.404	0.338	0.937
SEK	+0.62	+0.09	−3.15	−4.06	−1.00	−3.36	−5.92	+1.25
$NW_{12}[p]$			0.545	0.567	0.876	0.513	0.384	0.849
WCB[p]			0.588	0.644	0.903	0.577	0.464	0.864
GBP	+1.16	+0.04	−2.79	−2.37	−2.67	−1.48	−2.85	+0.90
$NW_{12}[p]$			0.554	0.689	0.714	0.753	0.621	0.905
WCB[p]			0.580	0.731	0.807	0.780	0.677	0.863
EUR	−0.21	+0.02	+1.45	+2.76	−0.83	+0.36	+1.73	−1.72
$NW_{12}[p]$			0.775	0.656	0.918	0.942	0.771	0.828
WCB[p]			0.786	0.690	0.883	0.945	0.792	0.807
CHF	−1.65	−0.06	+1.45	+3.18	−1.44	+0.43	+1.36	−1.03
$NW_{12}[p]$			0.748	0.486	0.867	0.924	0.758	0.906
WCB[p]			0.767	0.520	0.861	0.927	0.776	0.866
JPY	−2.18	−0.11	−0.39	+5.35	−8.59	−2.12	+2.98	−8.94
$NW_{12}[p]$			0.920	0.359	0.132	0.604	0.623	0.171
WCB[p]			0.935	0.486	0.045	0.655	0.693	0.022

3.2 Interest Rate Differentials and Currency Betas

Crashes in the carry trade happen when the high-interest rate currencies depreciate significantly more relative to the low-interest rate currencies. When a currency is offering a high interest rate, it compensates for the risk that it bears, including exposure to global growth.

This channel relates the currency IRD with the currency beta, and in Table 4, it is shown that the two are highly correlated across the sample ($\rho = 0.87$).

Connecting the IRD and beta to the IYC signals shows how they perform across different currencies. The currencies with higher beta and IRD, such as AUD, NZD, and CAD depreciate significantly more compared to the safe haven currencies (Ranaldo and Söderlind, 2010) with lower beta and IRD, such as the JPY, CHF, and EUR. Especially with the downcurve filter, the regression coefficients of these individual currencies goes from extremely negative for the AUD to positive for the JPY. The carry trade and beta HML portfolio long the high-beta high-IRD currencies while short the low-beta low-IRD currencies, which explains why those portfolios crash during the IYC signals. Figure 6 shows the relationship between the currency beta and IRD, and their performance during the signal months.

A further study on this can be conducted in a monthly panel by estimating the coefficient of the interaction between IYC and beta, and IYC and IRD:

$$\Delta s_{i,t+1} = \alpha_i + \gamma_1 \, \mathrm{IYC}_t + \gamma_2 \, \hat{\beta}_{i,t} + \gamma_3 \left(\mathrm{IYC}_t \times \hat{\beta}_{i,t} \right) + e_{i,t+1}, \tag{13}$$

$$\Delta s_{i,t+1} = \alpha_i + \gamma_1 \, \mathrm{IYC}_t + \gamma_2 \, \mathrm{IRD}_{i,t} + \gamma_3 \left(\mathrm{IYC}_t \times \mathrm{IRD}_{i,t} \right) + e_{i,t+1}, \tag{14}$$

The rationale of doing this exercise is to capture the signal effect on IRD or beta sorting in a portfolio setting. Both regressions include currency fixed effects α_i. I estimate them separately because their sample averages are highly correlated. Additionally, this exercise includes the exposure measure in real time ($\hat{\beta}_t$ and $\mathrm{IRD}_{i,t}$) and with the sample average ($\bar{\beta}_i$ and $\overline{\mathrm{IRD}_t}$). In the latter case, the fixed effects absorb the exposure's level, while its interaction with the signal captures the differences in the signal effect across the currencies. Because all currencies share the dollar factor, I use Driscoll and Kraay (1998) to account for that, in addition to the WCB ($B = 9,999$).

Table 5: Signal interaction with beta and IRD from (13) & (14). Column (i) provides the pooled result of the IYC signal across the G10 currencies without any interaction. Column (ii) & (iv) provide the results for the interaction with real-time beta and IRD. Column (iii) & (v) provide the results for the interaction with sample average beta and IRD. The reported p-values are Driscoll and Kraay (1998) (12-lag) and WCB($B = 9,999$), with the sample running from 1988:01–2026:02 (458 months).

	(i)	(ii)	(iii)	(iv)	(v)
IYC_t	−4.13	−3.82	−1.59	−3.48	−2.91
$\text{DK}_{12}[p]$	0.284	0.319	0.649	0.332	0.418
WCB$[p]$	0.352	0.413	0.685	0.440	0.478
$\hat{\beta}_{i,t}$		+9.04			
$\text{DK}_{12}[p]$		0.144			
WCB$[p]$		0.168			
$\text{IYC}_t \times \hat{\beta}_{i,t}$		+1.74			
$\text{DK}_{12}[p]$		0.909			
WCB$[p]$		0.885			
$\text{IYC}_t \times \bar{\beta}_i$			−42.42		
$\text{DK}_{12}[p]$			0.035		
WCB$[p]$			0.024		
$\text{IRD}_{i,t}$				+0.49	
$\text{DK}_{12}[p]$				0.593	
WCB$[p]$				0.623	
$\text{IYC}_t \times \text{IRD}_{i,t}$				−1.76	
$\text{DK}_{12}[p]$				0.190	
WCB$[p]$				0.330	
$\text{IYC}_t \times \overline{\text{IRD}}_i$					−2.80
$\text{DK}_{12}[p]$					0.020
WCB$[p]$					0.012
Currency fixed effects	Yes	Yes	Yes	Yes	Yes
n	4122	4122	4122	4122	4122

In Table 5, the signal coefficient is negative, but statistically insignificant (column (i)). It makes sense because in Table 4, only AUD and NZD are statistically significant for the baseline signal, while the rest are insignificant or even positive. The real-time interaction with beta and IRD also does not produce significant results, yielding positive coefficients instead. On the other hand, the interaction with the sample average beta and IRD is the one that produces a negative and statistically significant regression coefficients, with is a corollary of Table 4.

3.3 Fama Regressions

The results in Section 3.2 suggest an implication for the forward premium puzzle. There is a significant depreciation for currencies when the IYC is active. So, the question here is whether it can explain help to explain the puzzle by providing a period where it mean reverts. I estimate the Fama (1984) regression with the signals in order to study this. The first equation is the original regression, while the second one introduces the signal in the regression:

$$\Delta s_{i,t+1} \;=\; \alpha_i \;+\; \gamma_F \, \tfrac{1}{12} \left(i_{i,t} - i_{US,t}\right) \;+\; e_{i,t+1}, \tag{15}$$

$$\Delta s_{i,t+1} \;=\; \alpha_i \;+\; \gamma_F \, \tfrac{1}{12} \left(i_{i,t} - i_{US,t}\right) \times \text{Signal}_t \;+\; e_{i,t+1}, \tag{16}$$

If the UIP holds, the regression coefficient for (15) (γ_F) must be equal to -1. However, it is well documented in the literature and also in Table 6 that most of the currencies instead have a positive γ_F, with the pooled coefficient across G10 currencies equal to +0.49. After introducing the signals into the equation, the distinction is clear. We can interpret the result as a slow consistent deviation from the UIP ("staircase up") in the months that are downcurve IYC months, and a violent reversion ("elevator down") for the UIP during the non-downcurve IYC months in the terminology of Brunnermeier et al. (2008). This result also complements other studies on the puzzle (Verdelhan, 2010; Farhi and Gabaix, 2016; Chernov et al., 2018) by introducing a state variable for the mean reversion periods.

3.4 Tail or Distribution Shift?

A further question regarding the signals is whether they make the tail of the return distribution fatter or shift the overall return distribution. So, in Table 7 I present the $5^{th}, 10^{th}, 25^{th}, 50^{th}, 75^{th}$ percentile of the return distribution in addition to the annualized volatility and skewness for each of the portfolio for different signals. The fifth percentile is used as the proxy for left

Table 6: Fama regression incorporating the signals. The resulting regression follows (16), with reported p-values for each currency are Newey and West (1987) (12-lag) and WCB($B = 9,999$), while the pooled regression replaces Newey and West (1987) with Driscoll and Kraay (1998) (12-lag) to take into account the currency fixed effect. The sample runs from 1988:01–2026:02 (458 months).

	No Signal	IYC Signal	Non-Downcurve	Downcurve
Currency	γ_F	γ_F	γ_F	γ_F
AUD	+1.44	+0.26	−6.48	+0.33
$NW_{12}[p]$	0.112	0.834	0.223	0.677
WCB$[p]$	0.264	0.847	0.722	0.806
CAD	−0.87	+1.14	−4.56	−0.47
$NW_{12}[p]$	0.343	0.152	0.494	0.623
WCB$[p]$	0.532	0.286	0.738	1.000
CHF	+2.19	−2.06	−6.66	+0.40
$NW_{12}[p]$	0.035	0.399	0.020	0.916
WCB$[p]$	0.066	0.534	0.079	0.871
EUR	+1.66	−5.89	−9.61	−2.02
$NW_{12}[p]$	0.206	0.047	0.001	0.630
WCB$[p]$	0.152	0.447	0.180	0.968
GBP	+0.25	−0.79	−4.65	+0.56
$NW_{12}[p]$	0.881	0.641	0.252	0.818
WCB$[p]$	0.810	0.761	0.503	0.843
JPY	+1.10	+1.59	+2.57	+3.55
$NW_{12}[p]$	0.225	0.120	0.050	0.000
WCB$[p]$	0.302	0.400	0.367	0.124
NOK	−0.73	−3.43	−5.51	−2.35
$NW_{12}[p]$	0.544	0.185	0.182	0.412
WCB$[p]$	0.817	0.415	0.536	0.753
NZD	+0.88	−1.45	−6.50	+3.05
$NW_{12}[p]$	0.242	0.489	0.011	0.001
WCB$[p]$	0.449	0.683	0.206	0.154
SEK	−0.73	−1.76	−6.22	−1.00
$NW_{12}[p]$	0.616	0.283	0.200	0.587
WCB$[p]$	0.902	0.457	0.522	0.691
Pooled, Currency Fixed Effects: Implied Slope by State				
Pooled	+0.49	−1.27	−4.37	+0.34
WCB$[p]$		0.319	0.019	0.976
$DK_{12}[p]$		0.190	0.032	0.968

tail return, while the others are used as a proxy for the return distribution.

For the carry trade portfolio, the IYC signal with and without the downcurve filter behaves differently; In the months where the downcurve filter is applied, it shifts the fifth percentile to the left by almost 3%, while the other reported percentiles by around 1.5%. On the other hand, the IYC signal with the downcurve shape actually shifts the fifth percentile

Table 7: Distribution of the portfolio returns conditioning the signals. This table presents the $5^{th}, 10^{th}, 25^{th}, 50^{th}, 75^{th}$ percentiles of the portfolio returns across different signal states. Additionally, it also reports the annualized volatility and skewness of the returns. The sample runs from 1988:01–2026:02 (458 months).

Portfolio	State	n	q5	q10	q25	q50	q75	Vol (%/yr)	Skew
Carry, spot	Off	365	−3.97	−2.41	−0.97	0.44	1.55	7.44	−0.52
	On	93	−4.96	−3.79	−1.75	−0.75	0.74	8.94	−1.05
	On, non-downcurve	56	−6.80	−4.11	−2.39	−1.03	0.02	9.85	−1.10
	On, downcurve	37	−2.41	−1.71	−1.26	−0.17	1.37	6.68	0.27
Carry, total	Off	365	−3.69	−2.18	−0.65	0.69	1.89	7.48	−0.45
	On	93	−4.44	−3.38	−1.41	−0.16	1.19	8.98	−1.04
	On, non-downcurve	56	−6.36	−3.71	−2.00	−0.58	0.33	9.82	−1.11
	On, downcurve	37	−1.85	−1.35	−0.93	0.27	1.77	6.67	0.41
Beta HML, spot	Off	365	−3.65	−2.69	−1.03	0.30	1.78	8.33	0.04
	On	93	−5.87	−3.94	−1.84	−0.49	0.79	9.75	−0.98
	On, non-downcurve	56	−5.20	−3.93	−2.32	−0.89	0.15	9.10	−1.66
	On, downcurve	37	−6.11	−4.13	−1.20	−0.06	1.62	10.32	−0.58
Beta HML, total	Off	365	−3.47	−2.58	−0.97	0.46	1.92	8.38	0.07
	On	93	−5.49	−3.83	−1.68	−0.29	0.72	9.72	−0.92
	On, non-downcurve	56	−5.05	−3.80	−2.28	−0.91	0.30	9.06	−1.60
	On, downcurve	37	−5.67	−4.27	−1.21	0.02	1.67	10.38	−0.50

to the right, consistent with the high IRD compensation reducing the left tail returns of the distribution. The downcurve regime also shows a lower volatility and actually provides a positive skewness, while the non-downcurve regime explains most of the volatility and negative skewness of the carry trade.

Meanwhile, the beta HML portfolio has a different result than the carry trade portfolio; the downcurve filter does not have any effect within the IYC signal. The IYC signal as a whole shifts mostly the tail of the distribution (by around 2% to the left), while it shifts the other percentiles by around 1%. In fact, the downcurve months are the periods with highest beta HML volatility, while the non-downcurve months provide the most negative skewness. What we can learn from this is that the IYC signal affects high-beta currencies consistently across the board; while the high-interest-rate currencies do not depreciate significantly during the downcurve months. This result has implications for the divergence between high-beta and high-interest-rate currencies that can be explored in the future. Table B.6 provides an

analysis of the degree to which the signals capture the left tail returns.

3.5 Portfolio Trading and Risk Management Application

This section presents a real-life application of the signals in the carry trade portfolio following Section 2.3. I provide the simulated returns[5] of a fully observable signal at time t, so that the investor can take action on the carry trade portfolio at the end of time t for a realized return at $t+1$. For the first set of strategies, the carry trade portfolio will be fully unwind (no position) or reversed into anti-carry position. For the second set of strategies, I apply the downcurve overwrite (staying in even though we observe the IYC signal). Figure 7 shows the simulated returns in the full sample of these strategies, with the IYC signal and the downcurve filter shaded in the timeline. Table 8 provides the summary statistics (mean, Sharpe ratio, skewness, 5% expected shortfall, maximum drawdown, worst month return, and comparison with Lustig et al. (2011) benchmark for rebalancing the carry trade portfolio) of the carry trade and beta HML portfolios after applying the strategies.

The strategies improve the portfolio performance metrics across the board, without the downcurve filter, the carry trade (beta HML) portfolio increases the mean return from 4.19% (2.54%) to 4.93% (4.22%), Sharpe ratio from 0.53 (0.29) to 0.73 (0.56), and reduces the worst month return from -11.61% (-12.51%) to -7.27% (-6.55%). Meanwhile, applying the downcurve filter increases the mean return to 5.45% (4.16%), Sharpe ratio to 0.78 (0.52), and the worst month return to -7.27% (-7.64%). The improvement in carry trade portfolio is statistically significant at 10% after applying the downcurve filter, while the beta HML portfolio is statistically significant at 5% across the board. Similar to the return distribution in Section 3.4, the beta HML portfolio is relatively the same (or even slightly worse) after applying the downcurve filter. Implementing the anti-carry instead of unwinding the portfolio provides a higher return and Sharpe ratio, while fully removes the negative skewness (Brunnermeier et al., 2008) of the return. However, the anti-carry can be costly if the carry

[5]All returns in this paper are reported before transaction costs.

trade does not crash, for example during the late 1980s to early 1990s where the IYC signal coincides with the downcurve regime, the carry trade actually does well and implementing the anti-carry hurts the portfolio return.

The main difference between this return analysis and the regression analysis in Section 3.1 is that the analysis here replaces the monthly return with 0 without changing the number of sample months, while the regression analysis actually separates the signal and non-signal months.

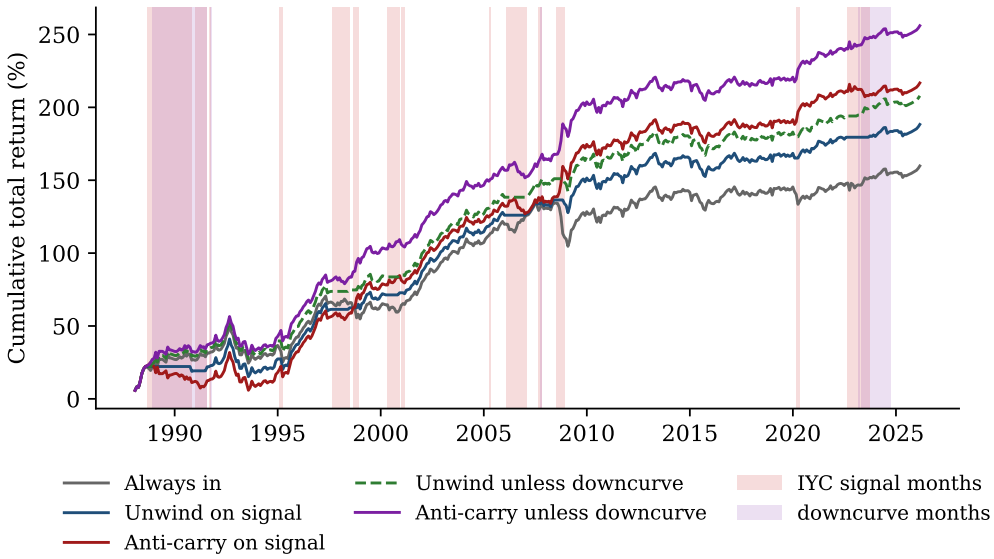


Figure 7: Trading the IYC and downcurve signals. This figure shows the cumulative carry trade returns after applying the signals. The months shaded with pink are the IYC signal months where the unwind or anti-carry applied to the portfolio. The months shaded with purple are the months with downcurve shape, where the overwrite of the signal is applied. The sample runs from 1988:01–2026:02 (458 months).

Table 8: Summary statistics for the portfolios after trading application of the signals. The performance metrics column presents the mean return, Sharpe ratio, skewness, 5% expected shortfall, maximum drawdown, worst month return for the baseline portfolio and its trading application of the signals. The next column shows the signals implementation improvement against the baseline portfolio. The reported p-values are Ledoit and Wolf (2008) and circular block-bootstrap (CBB, B=4,999) to measure the difference of the two series with the same length and mostly overlapping observations. The sample runs from 1988:01–2026:02 (458 months).

	Performance Metrics						Difference from Always In		
Strategy	Mean (%/yr)	Sharpe	Skew	$ES_{5\%}$ (%/mo)	Max DD (%)	Worst Month (%/mo)	ΔSR	ΔMean (%/yr)	ΔSkew
Panel A: The Carry Trade (Total Returns)									
Always in	+4.19	0.53	-0.68	-5.30	-30.1	-11.61			
Unwind on signal	+4.93	0.73	-0.34	-4.48	-25.9	-7.27	+0.20	+0.75	+0.35
LW[p]							0.071	0.293	
CBB[p]							0.122	0.292	0.164
Anti-carry on signal	+5.68	0.73	-0.02	-4.71	-25.9	-7.27	+0.20	+1.49	+0.66
LW[p]							0.287	0.293	
CBB[p]							0.318	0.292	0.139
Unwind unless downcurve	+5.45	0.78	-0.33	-4.51	-25.9	-7.27	+0.25	+1.26	+0.36
LW[p]							0.022	0.057	
CBB[p]							0.079	0.066	0.115
Anti-carry unless downcurve	+6.71	0.87	-0.01	-4.59	-25.9	-7.27	+0.34	+2.52	+0.68
LW[p]							0.051	0.057	
CBB[p]							0.094	0.066	0.119
Panel B: The Beta HML Portfolio (Total Returns)									
Always in	+2.54	0.29	-0.27	-5.62	-30.1	-12.51			
Unwind on signal	+4.22	0.56	+0.20	-4.45	-14.0	-6.55	+0.27	+1.68	+0.47
LW[p]							0.002	0.016	
CBB[p]							0.014	0.020	0.036
Anti-carry on signal	+5.90	0.68	+0.32	-4.73	-14.8	-6.55	+0.39	+3.37	+0.59
LW[p]							0.010	0.016	
CBB[p]							0.021	0.020	0.124
Unwind unless downcurve	+4.16	0.52	+0.02	-4.94	-22.1	-7.64	+0.23	+1.62	+0.29
LW[p]							0.002	0.003	
CBB[p]							0.020	0.005	0.178
Anti-carry unless downcurve	+5.79	0.67	+0.19	-4.97	-22.1	-7.64	+0.38	+3.25	+0.46
LW[p]							0.002	0.003	
CBB[p]							0.014	0.005	0.309

4 Horse Race Against Other Measures of Risk

The predictive regression (12) does the study without any controls. In this section, I extend the analysis to include variables that are known in the literature to gauge risk in the financial market. The generic form of the regression is the following:

$$\text{Return}_{t+1} = \alpha + \gamma \text{State}_t + \delta' W_t + e_{t+1} \tag{17}$$

Where W_t is a vector of control variable(s) for the study, and other specifications follow (12). Section 4.1 provides a horse race against other yield curve variables and Section 4.2 shows a horse race against volatility indices and credit risk.

4.1 Individual, United States, and Global Yield Curves

The United States yield curve is often used as a proxy for a global risk in the literature, which makes sense since the United States is the biggest economy in the world, and if something is happening there, it should have an impact to the rest of the world. Therefore, it is natural to question whether the IYC signal adds information beyond the United States yield curve. This is also the reason why I excluded the United States yield curve when constructing the baseline IYC signal. An alternative construction including the United States and the dollar does not change the result (Table B.5).

The IYC signal is constructed from the aggregation of the G10 yield curves. So, it is important to test it against the aggregate G10 yield slope to show that it adds information beyond the aggregate slope. Table 9 presents the horse race against the United States inversion, level and change of the 10Y-2Y slope, the level and change of the G10 10Y-2Y slope. The result is robust controlling all of these yield curve variables with none of those variables are statistically meaningful, which confirms that the IYC signal add more information than the United States and aggregate G10 yield curves.

Table 9: Horse race regression against the United States and aggregate G10 yield curves. The regression follows (17) with $\text{State}_t = \text{IYC}_t$. The columns with (1) provides the regression without control from (12), while (2) provides the horse race results. The slope definition uses 10Y-2Y slope, while the U.S. inversion is when the value is negative (57 months). The global slope uses the average of the G10 individual slope. The reported p-values are Newey and West (1987) (12-lag) and WCB($B = 9,999$), with the sample running from 1988:01–2026:02 (458 months).

	Carry Spot		Carry Total		Beta HML Spot		Beta HML Total	
	(1)	(2)	(1)	(2)	(1)	(2)	(1)	(2)
IYC signal	−11.63	−14.62	−9.86	−13.10	−12.82	−17.31	−13.58	−18.20
$\text{NW}_{12}[p]$	0.003	0.002	0.013	0.006	0.001	0.000	0.000	0.000
WCB$[p]$	0.001	0.003	0.007	0.020	0.006	0.005	0.001	0.002
Global slope		+3.39		+2.48		−1.73		−1.25
$\text{NW}_{12}[p]$		0.329		0.492		0.651		0.736
WCB$[p]$		0.311		0.528		0.775		0.823
Δ Global slope (1m)		−36.54		−35.31		−15.18		−14.97
$\text{NW}_{12}[p]$		0.062		0.071		0.477		0.479
WCB$[p]$		0.287		0.290		0.554		0.560
U.S. inversion		+2.75		+2.19		+1.06		+1.95
$\text{NW}_{12}[p]$		0.513		0.611		0.798		0.629
WCB$[p]$		0.616		0.692		0.847		0.719
U.S. slope		−3.07		−2.84		−2.01		−2.23
$\text{NW}_{12}[p]$		0.144		0.184		0.254		0.191
WCB$[p]$		0.185		0.227		0.424		0.365
Δ U.S. slope (1m)		+21.98		+20.99		+10.24		+10.08
$\text{NW}_{12}[p]$		0.073		0.086		0.443		0.450
WCB$[p]$		0.221		0.237		0.490		0.505
$\bar{R}^2$ (%)	2.7	4.2	1.9	3.1	2.7	2.5	3.0	2.9
n	458	458	458	458	458	458	458	458

Table B.10 provides an additional testing at the currency level against the information from individual country (own inversion, own slope level and change, and G10 slope level) and its relative slope against the United States (Chen and Tsang, 2013). The result holds qualitatively, with safe have currencies such as JPY, CHF, and EUR coefficient become positive, while the other currencies having a more negative coefficients.[6]

[6]Currency level inference is weak by construction, since each regression uses one currency and fourteen episodes. The evidence for the mechanism comes from the cross-sectional ranking of these coefficients thru beta and IRD (Section 3.2).

4.2 Global Risk and Currency Factors

The carry trade literature has documented many factors that explain the movement in the carry trade, both in a predictive and contemporaneous setting. Therefore, I test the signals with those variables to ensure that they are not just a proxy of already known global risk factors. Table 10 provides the regression coefficient in a horse race of predictive regression for the signals in the carry trade and beta HML portfolios. I run a ladder regression starting with the IYC signal by itself in column (i), adding change and level of NFCI as a proxy for financial condition in column (ii), adding the dollar factor and commodity return in column (iii), adding the change and level of FX volatility in column (iv), adding the change in CLI in column (v), adding the change and level of VIX in column (vi), and applying the downcurve filter in column (vi)'. The result for the carry trade is robust at 1% significant level before adding the VIX, and it is robust at 5% after the VIX addition; the only other variable that is significant is the change in NFCI. Meanwhile, the result for beta HML is robust at 1% significant level across the board, with no other variable is significant at 5% level. The complete table with each of coefficient for the controlling variable can be seen in Table B.8 and Table B.9.

5 Emerging Market Extensions

5.1 Emerging Market Currencies: Beta and IRD

Most of the carry trade literature focuses on the G10 currencies, because they are very liquid, highly traded, and the data is readily available. The EM currencies however, are actually very attractive for carry trade purposes, mainly due to their very high IRD. I selected seven EM currencies that have the highest trading volume and that are not pegged, to capture their actual movement. Table 11 presents the same study as in Table 4 and it shows many

Table 10: The simplified results of the carry trade and beta HML portfolios. The reported numbers are the regression coefficients of $\text{Return}_{t+1} = \alpha + \gamma\,\text{State}_t + \delta' W_t + e_{t+1}$ for both the spot only and total portfolio returns. Column (i) reports the result without any controls. Column (ii) adds a control for financial conditions using NFCI; column (iii) further adds a control for the dollar factor and commodity return; column (iv) takes into account the FX Volatility; column (v) includes CLI; column (vi) adds the VIX factors; and column (vi)' applies the downcurve filter. The reported p-values are Newey and West (1987) (12-lag) and WCB($B = 9,999$). The first five columns use a sample from 1988:01–2026:02 (458 months), while the last two columns use a sample from 1990:12–2026:02 (423 months) due to data availability for the VIX. The complete version of this study with coefficient for each variable is provided in Table B.8 and Table B.9.

	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(i)′
Carry, spot	−11.63	−10.60	−10.64	−11.82	−11.83	−11.55	−13.05
$\text{NW}_{12}[p]$	0.003	0.000	0.000	0.000	0.000	0.009	0.010
WCB$[p]$	0.001	0.001	0.000	0.000	0.000	0.024	0.004
Carry, total	−9.86	−9.04	−9.08	−10.40	−10.43	−10.66	−12.17
$\text{NW}_{12}[p]$	0.013	0.002	0.002	0.001	0.001	0.014	0.015
WCB$[p]$	0.007	0.003	0.003	0.003	0.003	0.037	0.009
Beta HML, spot	−12.82	−12.63	−12.59	−13.38	−13.41	−15.42	−17.07
$\text{NW}_{12}[p]$	0.001	0.000	0.001	0.000	0.000	0.000	0.000
WCB$[p]$	0.006	0.004	0.006	0.002	0.003	0.002	0.001
Beta HML, total	−13.58	−13.67	−13.63	−14.50	−14.52	−15.87	−17.33
$\text{NW}_{12}[p]$	0.000	0.000	0.000	0.000	0.000	0.000	0.000
WCB$[p]$	0.001	0.000	0.001	0.000	0.000	0.001	0.001
n	458	458	458	458	458	423	423

of them yields higher IRD compared to the G10 currencies. One of the reasons for their high IRD is due to the peso problem (Burnside et al., 2011) and other risks that these currencies have.

I use the signals constructed from the G10 yield curves and apply it to the EM currencies. The rationale behind this is that the G10 economies are relatively "safe" compared to the EM ones, as they do not have a peso problem and other elevated sovereign debt risks. They are also the ones that dictate the global financial cycle (Rey, 2013; Miranda-Agrippino and Rey, 2020) Therefore, one argument that we can make is that, the G10 IYC signal is a proxy for a global systematic risk in the financial market. One way to test this is to actually apply this signal to currencies outside the signal construction to see if the results hold. Table 11 shows that qualitatively the IYC signal works, but the sorting between IRD, beta, and the signal is not as clean as the one in G10. Table B.16 repeats the exercise from Table 5 to

Table 11: IRD and beta mechanism for EM currencies. This table repeats the study from Table 4 on EM currencies. The reported regression coefficient is $\Delta s_{i,t+1} = \alpha_i + \gamma_i \, \text{IYC}_t + e_{i,t+1}$. The reported p-values are Newey and West (1987) (12-lag) and WCB ($B = 9,999$) for the currency level, and Driscoll and Kraay (1998) (12-lag) replaces Newey and West (1987) in the pooled setting to take account the currency fixed effects. The sample runs from 1996:02–2026:02 (361 months) due to EM data availability.

Currency	IRD (%)	β	γ (%/yr)	$\text{NW}_{12}[p]$	WCB$[p]$
BRL	+11.75	+0.29	−6.62	0.449	0.479
CLP	+3.15	+0.17	−11.04	0.180	0.166
IDR	+7.40	+0.06	−28.22	0.246	0.547
INR	+4.87	+0.06	−7.25	0.050	0.145
MXN	+7.47	+0.30	−11.44	0.203	0.252
KRW	+1.54	+0.12	−17.10	0.111	0.170
ZAR	+5.89	+0.31	−15.39	0.059	0.024
Pooled, Currency Fixed Effects					
Pooled			−13.87		
WCB$[p]$			0.175		
$\text{DK}_{12}[p]$			0.082		

analyze the beta and IRD mechanism for the signal on EM currencies and the combined EM+G10 currencies. Corollary from Table 11, the mechanism results from the G10 does not hold on EM currencies and their combined portfolio. One possible explanation for this is that every EM currencies have an individual risk fixed effects that is baked in to their IRD to compensate that risk.

5.2 The Portfolio with Emerging Markets

In the portfolio setting, I combine the seven EM currencies with the G10 currencies to build the same 3×3 carry trade and beta portfolios. The idea is to create a portfolio that longs the EM currencies and funds them using the safe haven currencies (the EM currencies occupy 99% of the long-leg in the carry trade portfolio and 72% of the long-leg in the beta

Table 12: G10+EM summary statistics in signal and no-signal months. This table repeats the study from Table 3 (without the predictive regression) by adding EM currencies in *Panel B* and shorten the sample to 1996:02–2026:02 (361 months).

Portfolio	State	n	Mean (%/yr)	Sharpe	Skew	$\text{VaR}_{5\%}$ (%/mo)	$\text{ES}_{5\%}$ (%/mo)	Worst Month (%/mo)
Panel A: The G10 Portfolio								
Carry, spot	No signal	304	+2.63	0.39	−0.41	−3.35	−4.49	−6.04
	Signal on	57	−11.38	−1.13	−1.23	−6.80	−8.63	−12.09
Carry, total	No signal	304	+5.84	0.86	−0.39	−3.15	−4.21	−5.78
	Signal on	57	−7.06	−0.70	−1.22	−6.36	−8.27	−11.61
Beta HML, spot	No signal	304	+2.11	0.27	−0.26	−3.87	−4.98	−6.84
	Signal on	57	−12.01	−1.24	−1.30	−5.16	−8.60	−12.77
Beta HML, total	No signal	304	+3.78	0.48	−0.23	−3.68	−4.80	−6.55
	Signal on	57	−10.36	−1.07	−1.24	−5.02	−8.40	−12.51
Panel B: The G10+EM Portfolio (16 Currencies)								
Carry, spot	No signal	304	+0.28	0.03	−0.80	−5.26	−7.01	−19.25
	Signal on	57	−12.15	−0.61	−1.64	−10.77	−17.16	−26.18
Carry, total	No signal	304	+11.26	1.04	−0.33	−4.08	−5.73	−16.55
	Signal on	57	+3.75	0.19	−1.17	−9.35	−15.02	−23.43
Beta HML, spot	No signal	304	+0.85	0.09	−0.63	−5.03	−6.80	−12.53
	Signal on	57	−23.44	−1.26	−1.99	−11.79	−17.05	−25.92
Beta HML, total	No signal	304	+6.34	0.63	−0.49	−4.64	−6.23	−12.00
	Signal on	57	−14.85	−0.82	−1.68	−10.26	−15.95	−23.88

HML portfolio). The earliest available data for the full 16 currencies portfolio starting from February 1996, much later than the G10 only currencies due to EM data limitation. As a result, the IYC signal months is reduced to 57 months (compared to 93), with downcurve months significantly reduced to 7 months (compared to 36 months) because most of the downcurve occurs during high inflation periods from late 1980s to early 1990s when the EM data are not available.

Table 12 provides a summary statistics for the mean return, Sharpe ratio, skewness, 5% value at risk, 5% expected shortfall, and the worst month return for both G10 only and G10+EM portfolios in the same periods. The result from G10 portfolio holds qualitatively in the carry trade spot leg, and both beta HML spot leg and total return; the total return for the carry trade is actually positive on the signal months, because the IRD is so large that it compensate roughly a -12% annualized depreciation during those periods. This is another

Table 13: G10+EM signals predictive regression. This table presents the predictive regression of the signals constructed from G10 yield curves to portfolios of G10 and G10+EM. The reported coefficients are regressions of: $\text{Return}_{t+1} = \alpha + \gamma \text{State}_t^{G10} + e_{t+1}$. The reported p-values are Newey and West (1987) (12-lag) and WCB ($B = 9,999$), with the sample running from 1996:02–2026:02 (361 months).

	IYC Signal	Non-downcurve	Downcurve
Panel A: The G10 Portfolio			
Carry, spot	−14.01	−17.00	+8.68
$\text{NW}_{12}[p]$	0.015	0.006	0.010
WCB$[p]$	0.008	0.003	0.136
Carry, total	−12.91	−15.80	+8.90
$\text{NW}_{12}[p]$	0.023	0.010	0.010
WCB$[p]$	0.017	0.010	0.142
Beta HML, spot	−14.12	−17.17	+8.98
$\text{NW}_{12}[p]$	0.001	0.000	0.399
WCB$[p]$	0.003	0.000	0.430
Beta HML, total	−14.13	−17.32	+9.89
$\text{NW}_{12}[p]$	0.001	0.000	0.360
WCB$[p]$	0.003	0.000	0.313
Panel B: The G10+EM Portfolio			
Carry, spot	−12.43	−16.31	+15.45
$\text{NW}_{12}[p]$	0.132	0.047	0.002
WCB$[p]$	0.223	0.121	0.084
Carry, total	−7.51	−10.58	+13.86
$\text{NW}_{12}[p]$	0.281	0.137	0.003
WCB$[p]$	0.351	0.191	0.093
Beta HML, spot	−24.29	−28.35	+8.04
$\text{NW}_{12}[p]$	0.023	0.012	0.313
WCB$[p]$	0.019	0.014	0.333
Beta HML, total	−21.19	−24.46	+5.30
$\text{NW}_{12}[p]$	0.014	0.007	0.539
WCB$[p]$	0.014	0.018	0.548

application of the carry overwrite from Section 2.3, where the IRD fully covers the crash risk, and also another evidence that the IYC signal works better on the beta HML portfolio.

In the portfolio setting, I combine the seven EM currencies with the nine G10 and build the same three-long, three-short carry and beta portfolios of (**??**) over the combined sixteen. This shortens the sample to 1996:02–2026:02, for the reason given in Section 2.5. The resulting portfolio's long leg is dominated by the EM currencies, as their IRDs and betas imply, and they take 99% of the carry portfolio's long-leg slots and 72% of the beta HML portfolio's. Meanwhile, the funding legs are relatively unchanged, because no EM currency

here sits low enough on either the IRD or beta to enter them.

Table 13 presents the predictive regression of (12) on both G10 only and G10+EM portfolios for the same periods above. The result from G10 holds qualitatively to the extended portfolios, but with weaker power, especially when IRD is involved. The seven downcurve months coefficient for the carry trade are actually positive and are statistically significant even with small sample, which significantly affecting the results without the downcurve filter. Once again, the beta portfolios are very robust even after adding EM currencies with high IRDs.

6 Conclusion

This paper contributes to the yield curve and the international asset pricing literature by showing that the IYC signal constructed from clusters of fresh G10 yield curve inversions identify periods of elevated systematic crash risk for the currency market. The freshness of the inversion is important to distinguish information that still matters for the asset market and those that have gone stale; while the clustering helps to isolate systematic risk from idiosyncratic risk. The crashes live in high-beta and high-interest-rate currencies, which helps to predict the carry trade crashes. The signal covers about 20% of the sample, but is able to capture four of the five worst months in the carry trade. The result is robust upon controlling other global risk factors such as the VIX, FX Volatility, NFCI, and the United States yield curves and dollar factor.

This paper also provides an approach to distinguish different environment of the inversions, namely the growth-scare inversion and inflation-fighting inversion, by looking if the average G10 yield curve is strictly downward sloping ("*downcurve*). Inversions accompanied by this downcurve shape come during periods of high inflation, providing higher IRD for the carry trade, compensating most of the crash risk. An analogy to this can be seen when extending the study to EM currencies in general, where their spot is depreciating, but their

IRD is very high that it still provides a positive carry return. This result gives a perspective that some risks are compensated that it is worth bearing them.

Lastly, much of the existing literature evaluates the currency predictability over several horizons, which does not map directly into investors' portfolio decision. This paper contributes to the risk management literature by providing an actionable signal that investors can use in real-time to manage their currency portfolios. Investors can unwind partially or fully, go anti-carry, or hedge using options when the IYC signal is active. Moreover, when they also observe the downcurve shape, the can overwrite the signal and choose to stay invested because the risk that they are bearing compensated by the environment.

References

Accominotti, O., Cen, J., Chambers, D., and Marsh, I. W. (2019). Currency regimes and the carry trade. *Journal of Financial and Quantitative Analysis*, 54(5), 2233–2260.

Adrian, T., Crump, R. K., and Moench, E. (2013). Pricing the term structure with linear regressions. *Journal of Financial Economics*, 110(1), 110–138.

Andrews, S., Colacito, R., Croce, M. M., and Gavazzoni, F. (2024). Concealed carry. *Journal of Financial Economics*, 159, 103874.

Ang, A. and Chen, J. S. (2010). Yield curve predictors of foreign exchange returns. Working paper, Columbia Business School.

Backus, D. K., Foresi, S., and Telmer, C. I. (2001). Affine term structure models and the forward premium anomaly. *Journal of Finance*, 56(1), 279–304.

Bakshi, G. and Panayotov, G. (2013). Predictability of currency carry trades and asset pricing implications. *Journal of Financial Economics*, 110(1), 139–163.

Bansal, R. and Dahlquist, M. (2000). The forward premium puzzle: Different tales from developed and emerging economies. *Journal of International Economics*, 51(1), 115–144.

Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3), 823–866.

Barroso, P. and Santa-Clara, P. (2015). Beyond the carry trade: Optimal currency portfolios. *Journal of Financial and Quantitative Analysis*, 50(5), 1037–1056.

Bauer, M. D. and Mertens, T. M. (2018). Economic forecasts with the yield curve. *FRBSF*

Economic Letter, 2018-07.

Brunnermeier, M. K., Nagel, S., and Pedersen, L. H. (2008). Carry trades and currency crashes. *NBER Macroeconomics Annual*, 23, 313–348.

Burnside, C., Eichenbaum, M., Kleshchelski, I., and Rebelo, S. (2011). Do peso problems explain the returns to the carry trade? *Review of Financial Studies*, 24(3), 853–891.

Caballero, R. J. and Doyle, J. B. (2012). Carry trade and systemic risk: Why are FX options so cheap? NBER Working Paper 18644.

Cameron, A. C., Gelbach, J. B., and Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3), 414–427.

Campbell, J. Y. and Shiller, R. J. (1991). Yield spreads and interest rate movements: A bird's eye view. *Review of Economic Studies*, 58(3), 495–514.

Chen, Y.-C. and Tsang, K. P. (2013). What does the yield curve tell us about exchange rate predictability? *Review of Economics and Statistics*, 95(1), 185–205.

Chernov, M., Graveline, J., and Zviadadze, I. (2018). Crash risk in currency returns. *Journal of Financial and Quantitative Analysis*, 53(1), 137–170.

Christiansen, C., Ranaldo, A., and Söderlind, P. (2011). The time-varying systematic risk of carry trade strategies. *Journal of Financial and Quantitative Analysis*, 46(4), 1107–1125.

Clarida, R. and Waldman, D. (2008). Is bad news about inflation good news for the exchange rate? In *Asset Prices and Monetary Policy*, University of Chicago Press.

Clarida, R., Davis, J., and Pedersen, N. (2009). Currency carry trade regimes: Beyond the Fama regression. *Journal of International Money and Finance*, 28(8), 1375–1389.

Colacito, R., Riddiough, S. J., and Sarno, L. (2020). Business cycles and currency returns. *Journal of Financial Economics*, 137(3), 659–678.

Davidson, R. and MacKinnon, J. G. (2000). Bootstrap tests: How many bootstraps? *Econometric Reviews*, 19(1), 55–68.

Della Corte, P., Riddiough, S. J., and Sarno, L. (2016). Currency premia and global imbalances. *Review of Financial Studies*, 29(8), 2161–2193.

Diebold, F. X., Li, C., and Yue, V. Z. (2008). Global yield curve dynamics and interactions: A dynamic Nelson–Siegel approach. *Journal of Econometrics*, 146(2), 351–363.

Dreher, F., Gräb, J., and Kostka, T. (2020). From carry trades to curvy trades. *The World Economy*, 43(3), 758–780.

Driscoll, J. C. and Kraay, A. C. (1998). Consistent covariance matrix estimation with spatially dependent panel data. *Review of Economics and Statistics*, 80(4), 549–560.

Du, W., Tepper, A., and Verdelhan, A. (2018). Deviations from covered interest rate parity.

Journal of Finance, 73(3), 915–957.

Engel, C. (2016). Exchange rates, interest rates, and the risk premium. *American Economic Review*, 106(2), 436–474.

Engstrom, E. C. and Sharpe, S. A. (2019). The near-term forward yield spread as a leading indicator: A less distorted mirror. *Financial Analysts Journal*, 75(4), 37–49.

Estrella, A. and Hardouvelis, G. A. (1991). The term structure as a predictor of real economic activity. *Journal of Finance*, 46(2), 555–576.

Estrella, A. and Mishkin, F. S. (1997). The predictive power of the term structure of interest rates in Europe and the United States: Implications for the European Central Bank. *European Economic Review*, 41(7), 1375–1401.

Estrella, A. and Mishkin, F. S. (1998). Predicting U.S. recessions: Financial variables as leading indicators. *Review of Economics and Statistics*, 80(1), 45–61.

Fama, E. F. (1984). Forward and spot exchange rates. *Journal of Monetary Economics*, 14(3), 319–338.

Farhi, E. and Gabaix, X. (2016). Rare disasters and exchange rates. *Quarterly Journal of Economics*, 131(1), 1–52.

Farhi, E., Fraiberger, S. P., Gabaix, X., Rancière, R., and Verdelhan, A. (2015). Crash risk in currency markets. NBER Working Paper 15062.

Gabaix, X. and Maggiori, M. (2015). International liquidity and exchange rate dynamics. *Quarterly Journal of Economics*, 130(3), 1369–1420.

Hansen, L. P. and Hodrick, R. J. (1980). Forward exchange rates as optimal predictors of future spot rates: An econometric analysis. *Journal of Political Economy*, 88(5), 829–853.

Harvey, C. R. (1988). The real term structure and consumption growth. *Journal of Financial Economics*, 22(2), 305–333.

Harvey, C. R. (1991). The term structure and world economic growth. *Journal of Fixed Income*, 1(1), 7–19.

Hassan, T. A. (2013). Country size, currency unions, and international asset returns. *Journal of Finance*, 68(6), 2269–2308.

Hassan, T. A. and Mano, R. C. (2019). Forward and spot exchange rates in a multi-currency world. *Quarterly Journal of Economics*, 134(1), 397–450.

He, Z., Kelly, B., and Manela, A. (2017). Intermediary asset pricing: New evidence from many asset classes. *Journal of Financial Economics*, 126(1), 1–35.

Ilzetzki, E., Reinhart, C. M., and Rogoff, K. S. (2019). Exchange arrangements entering the twenty-first century: Which anchor will hold? *Quarterly Journal of Economics*, 134(2), 599–646.

Jotikasthira, C., Le, A., and Lundblad, C. (2015). Why do term structures in different currencies co-move? *Journal of Financial Economics*, 115(1), 58–83.

Jurek, J. W. (2014). Crash-neutral currency carry trades. *Journal of Financial Economics*, 113(3), 325–347.

Kaebi, M. M. and Ferreira Batista Martins, I. (2025). Long-run interest rate differentials and the profitability of currency carry. Örebro University Working Paper 10/2025.

Koijen, R. S. J., Moskowitz, T. J., Pedersen, L. H., and Vrugt, E. B. (2018). Carry. *Journal of Financial Economics*, 127(2), 197–225.

Kremens, L. (2024). Positioning risk. Working paper, University of Washington.

Kremens, L. and Martin, I. (2019). The quanto theory of exchange rates. *American Economic Review*, 109(3), 810–843.

Ledoit, O. and Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.

Lettau, M., Maggiori, M., and Weber, M. (2014). Conditional risk premia in currency markets and other asset classes. *Journal of Financial Economics*, 114(2), 197–225.

Lloyd, S. P. and Marin, E. A. (2024). Exchange rates, yield curves and transitory risk. Working paper, November 2024. Previously circulated as "Exchange-Rate Risk and Business Cycles."

Lustig, H. and Verdelhan, A. (2007). The cross section of foreign currency risk premia and consumption growth risk. *American Economic Review*, 97(1), 89–117.

Lustig, H., Roussanov, N., and Verdelhan, A. (2011). Common risk factors in currency markets. *Review of Financial Studies*, 24(11), 3731–3777.

Lustig, H., Roussanov, N., and Verdelhan, A. (2014). Countercyclical currency risk premia. *Journal of Financial Economics*, 111(3), 527–553.

Lustig, H., Stathopoulos, A., and Verdelhan, A. (2019). The term structure of currency carry trade risk premia. *American Economic Review*, 109(12), 4142–4177.

MacKinnon, J. G., Nielsen, M. Ø., and Webb, M. D. (2023). Cluster-robust inference: A guide to empirical practice. *Journal of Econometrics*, 232(2), 272–299.

Menkhoff, L., Sarno, L., Schmeling, M., and Schrimpf, A. (2012). Carry trades and global foreign exchange volatility. *Journal of Finance*, 67(2), 681–718.

Miranda-Agrippino, S. and Rey, H. (2020). U.S. monetary policy and the global financial cycle. *Review of Economic Studies*, 87(6), 2754–2776.

Mönch, E. (2012). Term structure surprises: The predictive content of curvature, level, and slope. *Journal of Applied Econometrics*, 27(4), 574–602.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.

Ranaldo, A. and Söderlind, P. (2010). Safe haven currencies. *Review of Finance*, 14(3), 385–407.

Ready, R., Roussanov, N., and Ward, C. (2017). Commodity trade and the carry trade: A tale of two countries. *Journal of Finance*, 72(6), 2629–2684.

Rey, H. (2013). Dilemma not trilemma: The global financial cycle and monetary policy independence. In *Global Dimensions of Unconventional Monetary Policy*, Jackson Hole Economic Policy Symposium, Federal Reserve Bank of Kansas City.

Richmond, R. J. (2019). Trade network centrality and currency risk premia. *Journal of Finance*, 74(3), 1315–1361.

Rietz, T. A. (1988). The equity risk premium: A solution. *Journal of Monetary Economics*, 22(1), 117–131.

Verdelhan, A. (2010). A habit-based explanation of the exchange rate risk premium. *Journal of Finance*, 65(1), 123–146.

Verdelhan, A. (2018). The share of systematic variation in bilateral exchange rates. *Journal of Finance*, 73(1), 375–418.

Wachter, J. A. (2013). Can time-varying risk of rare disasters explain aggregate stock market volatility? *Journal of Finance*, 68(3), 987–1035.

Wright, J. H. (2011). Term premia and inflation uncertainty: Empirical evidence from an international panel dataset. *American Economic Review*, 101(4), 1514–1534.

Wu, J. C., and Xia, F. D. (2016). Measuring the macroeconomic impact of monetary policy at the zero lower bound. *Journal of Money, Credit and Banking*, 48(2–3), 253–291.

A Economic Regimes

Table A reports the stylized facts across different economic regimes aggregated across the G10 currencies. The sample starts from January 1988 during the Great Moderation and Globalization regime and it covers significant economics eras such as the Dot-com bubble, the Great Recession, Zero Lower Bound (ZLB), COVID-19, and Quantitative Easing driven post-COVID inflation. The carry trade IRD and return vary across these regime, which is well documented in the literature. The downcurve signals are active during the high inflation

era, while ZLB era has no signal due to the relatively low spread between the maturities and also flat yield curve; similar reason why Japan's yield curve is never inverted during their negative interest rate regime. Potential extension using the shadow yield curve (Wu and Xia, 2016) can be explored in the future.

Table A: Economic regimes across the sample. For each regime, the table reports the number of months, the average G10 three-month bond yield (%), the average G10 CPI YoY inflation (%), the annualized carry trade IRD (%), and the number of active IYC signal and downcurve months within each regime. The sample runs from 1988:01–2026:02 (458 months).

Regime	n	3M Yield	CPI YoY	IRD	Total Return	IYC Months	Downcurve Months
Inflation targeting (1988–94)	84	8.52	3.73	+5.94	+5.24	34	33
Great Moderation and dot-com (1995–2000)	72	4.52	1.44	+4.86	+4.59	23	0
Pre-GFC (2001–07)	84	3.41	1.66	+4.42	+9.56	16	1
GFC (2008–09)	24	2.66	2.03	+3.85	−1.36	5	0
ZLB (2010–19)	120	0.80	1.39	+2.49	+1.70	0	0
COVID-19 (2020–21)	24	−0.05	1.25	+0.94	−0.17	2	0
Post-COVID QE inflation (2022–26)	50	2.60	4.02	+3.11	+3.54	13	18

B Robustness

This appendix does robustness check for the signal construction, comparison with competing variables, and subsample stability.

B.1 Signal Construction

This section of the appendix focuses on performing the robustness check for signal construction. Table B.1 presents the signal construction using 10Y-3M slopes instead of 10Y-2Y slopes; the numbers of active signal month are relatively the same with roughly half of them overlapping with each other. The regression coefficients are qualitatively stable, with slightly lower significance. One potential explanation of the difference between the two is that the three-month bond yields are a function of the very short-term expectation policy rate, so there might be a lower or upper bound to it.

Table B.1: Slope-definition robustness check: rebuilding the signal using 10Y-3M slopes instead of 10Y-2Y slopes.

Slope Definition	n	Carry Spot	Carry Total	Beta HML Spot	Beta HML Total
10Y−2Y (baseline)	93	−11.63	−9.86	−12.82	−13.58
$NW_{12}[p]$		0.003	0.013	0.001	0.000
WCB[p]		0.001	0.007	0.006	0.001
10Y−3M	104	−7.46	−6.73	−8.41	−8.89
$NW_{12}[p]$		0.054	0.081	0.038	0.024
WCB[p]		0.035	0.062	0.024	0.012
10Y−2Y, excl. downcurve	56	−17.35	−16.51	−17.93	−18.03
$NW_{12}[p]$		0.002	0.003	0.000	0.000
WCB[p]		0.002	0.004	0.001	0.001
10Y−3M, excl. downcurve	69	−9.66	−9.94	−10.10	−10.23
$NW_{12}[p]$		0.062	0.048	0.031	0.025
WCB[p]		0.038	0.036	0.023	0.018

Table B.2: Cluster size robustness check: this table repeats the study of the IYC signal from Table 3 ($\text{Return}_{t+1} = \alpha + \gamma\,\text{IYC}_t + e_{t+1}$) by changing the threshold of the IYC activation to require 1, 3, or 4 countries; while also providing an exact numbers of country to explain the distinction between idiosyncratic and systematic risk. The reported p-values are Newey and West (1987) (12-lag) and WCB($B = 9,999$), with the sample running from 1988:01–2026:02 (458 months).

Cluster Size	n	Carry Spot	Carry Total	Beta HML Spot	Beta HML Total
≥ 1 (any inversion)	169	−4.02	−2.65	−9.52	−9.96
$NW_{12}[p]$		0.125	0.308	0.000	0.000
WCB[p]		0.155	0.374	0.003	0.001
$= 1$ (exactly one)	76	+6.84	+7.08	−1.03	−0.88
$NW_{12}[p]$		0.003	0.002	0.712	0.751
WCB[p]		0.014	0.021	0.762	0.797
≥ 2 (baseline)	93	−11.63	−9.86	−12.82	−13.58
$NW_{12}[p]$		0.003	0.013	0.001	0.000
WCB[p]		0.001	0.007	0.006	0.001
$= 2$ (exactly two)	41	−15.16	−14.23	−16.82	−17.16
$NW_{12}[p]$		0.006	0.011	0.002	0.001
WCB[p]		0.010	0.018	0.003	0.002
≥ 3	52	−6.43	−4.33	−6.99	−7.93
$NW_{12}[p]$		0.041	0.184	0.072	0.031
WCB[p]		0.116	0.264	0.182	0.081
$= 3$ (exactly three)	22	−11.63	−10.45	−8.98	−9.49
$NW_{12}[p]$		0.064	0.089	0.010	0.004
WCB[p]		0.197	0.245	0.065	0.032
≥ 4	30	−1.89	+0.68	−4.78	−5.96
$NW_{12}[p]$		0.548	0.845	0.426	0.300
WCB[p]		0.320	0.862	0.465	0.315

Table B.2 presents the robustness check for different number of thresholds for the cluster definition. I provide the results for $N \geq 1, 2, 3, 4$ to see if the threshold matters. Changing

the thresholds to 1 or 3 does not change the result qualitatively, only reduces the significance statistically. However, changing it to 4 does impact the result for the carry trade. This might sound counterintuitive, as more countries being inverted might imply that there is more risk, but requiring cluster size of 4 tends to happen later in the cycle where the information might have alredy been stale; there is a gradual increase in the size of the cluster as a function of time, and the main paper also documents the importance of the freshness of the signal, which has the implication in this case. Additionally, I provide the results for exact number of cluster ($N = 1, 2, 3$) in order show that there is a big distinction between idiosyncratic risk and systematic risk. When there is exactly one inversion among the countries, the regression coefficients for the carry trade are actually positive and statistically significant, which ironically are good months for the carry trade. However, when we have exactly two and three countries that have inversions, the result changes drastically, emphasizing the distinction between idiosyncratic and systematic risk.

Table B.3: IYC deactivation robustness check: this table repeats the study of the IYC signal from Table 3 ($\text{Return}_{t+1} = \alpha + \gamma\,\text{IYC}_t + e_{t+1}$) by changing the threshold of the IYC deactivation, via slope steepening, to 1 and 3 consecutive months. The reported p-values are Newey and West (1987) (12-lag) and WCB($B = 9,999$), with the sample running from 1988:01–2026:02 (458 months).

Deactivation Rule	n	Carry Spot	Carry Total	Beta HML Spot	Beta HML Total
1 steepening month	40	−2.94	−0.81	−5.78	−6.64
$\text{NW}_{12}[p]$		0.436	0.837	0.076	0.039
WCB$[p]$		0.416	0.843	0.114	0.074
2 steepening months (baseline)	93	−11.63	−9.86	−12.82	−13.58
$\text{NW}_{12}[p]$		0.003	0.013	0.001	0.000
WCB$[p]$		0.001	0.007	0.006	0.001
3 steepening months	178	−5.31	−3.90	−5.67	−5.80
$\text{NW}_{12}[p]$		0.075	0.190	0.019	0.015
WCB$[p]$		0.061	0.221	0.026	0.013

Table B.3 presents the results by changing the threshold for consecutive slope steepening to 1 and 3. Requiring only one steepening month, would induce some noise in addition to make the signal deactivates too fast, while requiring three steepening months causes the signal to be active even longer than the naive count inversion, which defeats the purpose of

capturing the freshness of the signal. Nevertheless, the results are qualitatively robust.

Table B.4: Robustness check for different number of pairs in the portfolios. This table repeats the study of the IYC signal from Table 3 ($\text{Return}_{t+1} = \alpha + \gamma\,\text{IYC}_t + e_{t+1}$) by constructing 1×1, 2×2, 3×3, and 4×4 portfolios. The reported p-values are Newey and West (1987) (12-lag) and WCB($B = 9,999$), with the sample running from 1988:01–2026:02 (458 months).

		1×1	2×2	3×3	4×4
Carry, spot	IYC signal	−10.02	−11.00	−11.63	−10.14
	$\text{NW}_{12}[p]$	0.085	0.016	0.003	0.001
	WCB$[p]$	0.088	0.005	0.001	0.001
	non-downcurve	−19.51	−18.91	−17.35	−14.70
	$\text{NW}_{12}[p]$	0.019	0.006	0.002	0.001
	WCB$[p]$	0.024	0.005	0.002	0.001
	downcurve	+6.37	+3.37	−0.29	−0.86
	$\text{NW}_{12}[p]$	0.291	0.301	0.918	0.694
	WCB$[p]$	0.488	0.241	0.905	0.858
Carry, total	IYC signal	−7.74	−8.97	−9.86	−8.77
	$\text{NW}_{12}[p]$	0.200	0.059	0.013	0.007
	WCB$[p]$	0.276	0.112	0.007	0.005
	non-downcurve	−18.17	−17.76	−16.51	−14.07
	$\text{NW}_{12}[p]$	0.029	0.010	0.003	0.002
	WCB$[p]$	0.047	0.014	0.004	0.001
	downcurve	+9.39	+6.10	+2.37	+1.23
	$\text{NW}_{12}[p]$	0.153	0.081	0.351	0.526
	WCB$[p]$	0.497	0.273	0.396	0.594
Beta HML, spot	IYC signal	−13.22	−15.32	−12.82	−10.73
	$\text{NW}_{12}[p]$	0.013	0.001	0.001	0.002
	WCB$[p]$	0.051	0.011	0.006	0.026
	non-downcurve	−21.49	−22.40	−17.93	−15.27
	$\text{NW}_{12}[p]$	0.001	0.000	0.000	0.000
	WCB$[p]$	0.007	0.001	0.001	0.004
	downcurve	+2.26	−1.00	−2.01	−1.32
	$\text{NW}_{12}[p]$	0.754	0.857	0.740	0.810
	WCB$[p]$	0.741	0.892	0.858	0.815
Beta HML, total	IYC signal	−14.87	−16.29	−13.58	−11.44
	$\text{NW}_{12}[p]$	0.004	0.000	0.000	0.001
	WCB$[p]$	0.009	0.001	0.001	0.009
	non-downcurve	−21.70	−22.22	−18.03	−15.42
	$\text{NW}_{12}[p]$	0.001	0.000	0.000	0.000
	WCB$[p]$	0.007	0.001	0.001	0.003
	downcurve	−1.05	−3.38	−3.53	−2.65
	$\text{NW}_{12}[p]$	0.892	0.548	0.567	0.633
	WCB$[p]$	0.931	0.821	0.864	0.788

Table B.4 presents a robustness check for the number of pairs for the carry trade and beta HML portfolios, providing the results for $k = 1, 2, 3, 4$. The results hold qualitatively

across the board.

Table B.5 conducts a study from Table 3 by adding the United States yield curve when construction the IYC signal in addition to adding the United States dollar (USD) in the portfolios. The number of months with active IYC signal increases from 93 to 99 months, but the results are robust with very similar coefficients and p-values.

Table B.5: United States robustness check: this table repeats the study of the IYC signal from Table 3 ($\text{Return}_{t+1} = \alpha + \gamma\,\text{IYC}_t + e_{t+1}$) by adding the United States yield curve when construction the IYC and including the USD in the portfolios. The reported p-values are Newey and West (1987) (12-lag) and WCB($B = 9,999$), with the sample running from 1988:01–2026:02 (458 months).

Portfolio	Signal	Leg	γ (%/yr)	$\text{NW}_{12}[p]$	WCB[p]
G10 carry	Baseline (9 curves)	spot	−11.63	0.003	0.001
G10 carry	Baseline (9 curves)	total	−9.86	0.013	0.007
G10 carry	+US curve (10)	spot	−11.11	0.003	0.001
G10 carry	+US curve (10)	total	−9.43	0.013	0.008
Dollar-incl. carry	Baseline (9 curves)	spot	−11.83	0.007	0.000
Dollar-incl. carry	Baseline (9 curves)	total	−10.04	0.026	0.016
Dollar-incl. carry	+US curve (10)	spot	−11.56	0.006	0.000
Dollar-incl. carry	+US curve (10)	total	−9.87	0.023	0.012
G10 beta HML	Baseline (9 curves)	spot	−12.82	0.001	0.006
G10 beta HML	Baseline (9 curves)	total	−13.58	0.000	0.001
G10 beta HML	+US curve (10)	spot	−12.49	0.001	0.005
G10 beta HML	+US curve (10)	total	−13.18	0.000	0.001
Dollar-incl. beta HML	Baseline (9 curves)	spot	−11.24	0.004	0.036
Dollar-incl. beta HML	Baseline (9 curves)	total	−11.93	0.002	0.010
Dollar-incl. beta HML	+US curve (10)	spot	−10.84	0.005	0.030
Dollar-incl. beta HML	+US curve (10)	total	−11.47	0.002	0.008

B.2 The Naive Count Benchmark

This section of the appendix compares the IYC signal with the simplest construction for the clusters of inversions, the *naive count*. The naive count is active around 36% of the sample, nearly twice as much as the IYC signal's 20%. Figure B.1 provides a visual representation of when the naive count and the IYC signal are active. It can be seen that the naive count can be very persistent, active for years, which once again raises the concern whether the information provided there is still relevant or not.

Table B.6 presents the tail analysis for both naive count and the IYC signal. The

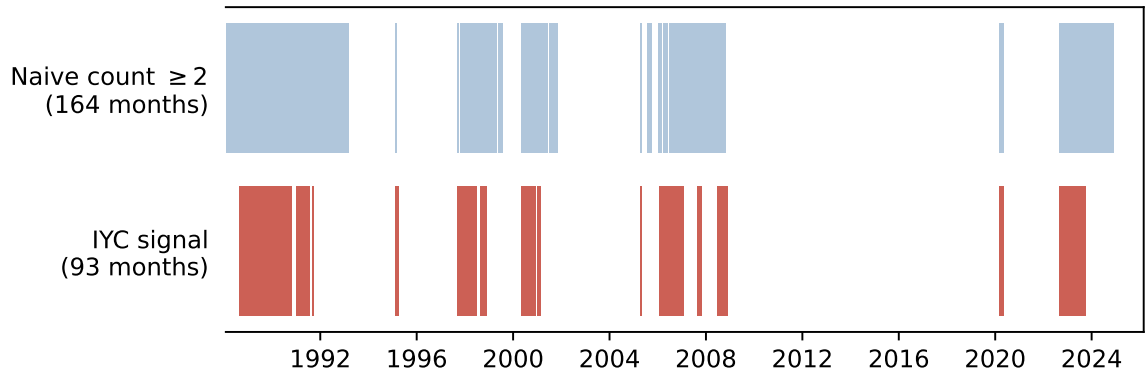


Figure B.1: IYC against the naive signal. The naive count (without any freshness filter, is active in 164 months) is shown against the IYC signal (with slope steepening freshness filter, is active in 93 months). The sample is from 1988:01–2026:02 (458 months).

Concentration measures the ratio of the number of tail events captured (using bottom 1%, 5%, and 10% as the proxy) and how often the signal is active in %. The In-State Rate measures the ratio between the number of tail events captured and number of months where the signal is active; the benchmark for this would be the 1%, 5%, and 10% used as the proxy for the tail. It shows that the IYC signal outperforms the threshold across the board using these metrics, while the naive count, although it captures more of the bottom 5% and 10% months, the accuracy is not as good, as the signal is diluted with stale information.

The main paper provides one way to capture the freshness of the signal by utilizing the steepening of the yield curve. Table B.7 provides an alternative way to capture the freshness of inversions using the naive count. By imposing a penalty $1/Age$ for naive count, it reduces the weight of the signal as it persists. It yields similar results to the IYC signal for both the carry trade and beta HML portfolios. This is another evidence that most of the crashes occur soon after the signal is active and the information becomes stale as it stays inverted.

Table B.6: How good are the IYC signal and naive count in capturing tail events. Concentration = $\frac{\%\text{ of tail captured}}{\%\text{ of active signal months}}$ and In-State Rate = $\frac{\#\text{ of tail months captured}}{\#\text{ of active signal months}}$. The bottom 1%, 5%, and 10% of the returns across the sample are used as a proxy for tail events. Panel A presents the results for carry trade portfolio and Panel B presents the results for beta HML portfolio. The sample is from 1988:01–2026:02 (458 months).

Panel A: Carry Portfolio

Tail Definition / State	n	% Active	Tail Captured	Concentration	In-State Rate
Worst 1% of Carry-Total Months (5 Months $\leq -6.03\%$/month)					
Naive count ≥ 2	164	36%	3/5 (60%)	1.68×	1.8%
Naive count excl. downcurve	112	24%	3/5 (60%)	2.45×	2.7%
IYC signal	93	20%	4/5 (80%)	3.94×	4.3%
IYC signal excl. downcurve	56	12%	4/5 (80%)	6.54×	7.1%
Worst 5% of Carry-Total Months (23 Months $\leq -3.75\%$/month)					
Naive count ≥ 2	164	36%	10/23 (43%)	1.21×	6.1%
Naive count excl. downcurve	112	24%	8/23 (35%)	1.42×	7.1%
IYC signal	93	20%	6/23 (26%)	1.28×	6.5%
IYC signal excl. downcurve	56	12%	5/23 (22%)	1.78×	8.9%
Worst 10% of Carry-Total Months (46 Months $\leq -2.32\%$/month)					
Naive count ≥ 2	164	36%	20/46 (43%)	1.21×	12.2%
Naive count excl. downcurve	112	24%	17/46 (37%)	1.51×	15.2%
IYC signal	93	20%	13/46 (28%)	1.39×	14.0%
IYC signal excl. downcurve	56	12%	12/46 (26%)	2.13×	21.4%

Panel B: Beta HML Portfolio

Tail Definition / State	n	% Active	Tail Captured	Concentration	In-State Rate
Worst 1% of Beta-HML-Spot Months (5 Months $\leq -6.53\%$/month)					
Naive count ≥ 2	164	36%	2/5 (40%)	1.12×	1.2%
Naive count excl. downcurve	112	24%	1/5 (20%)	0.82×	0.9%
IYC signal	93	20%	3/5 (60%)	2.95×	3.2%
IYC signal excl. downcurve	56	12%	2/5 (40%)	3.27×	3.6%
Worst 5% of Beta-HML-Spot Months (23 Months $\leq -4.16\%$/month)					
Naive count ≥ 2	164	36%	12/23 (52%)	1.46×	7.3%
Naive count excl. downcurve	112	24%	7/23 (30%)	1.24×	6.2%
IYC signal	93	20%	9/23 (39%)	1.93×	9.7%
IYC signal excl. downcurve	56	12%	5/23 (22%)	1.78×	8.9%
Worst 10% of Beta-HML-Spot Months (46 Months $\leq -2.84\%$/month)					
Naive count ≥ 2	164	36%	19/46 (41%)	1.15×	11.6%
Naive count excl. downcurve	112	24%	13/46 (28%)	1.16×	11.6%
IYC signal	93	20%	14/46 (30%)	1.50×	15.1%
IYC signal excl. downcurve	56	12%	9/46 (20%)	1.60×	16.1%

Table B.7: Freshness filter for the cluster of inversions. This table repeats the study of the from Table 3 ($\text{Return}_{t+1} = \alpha + \gamma\,\text{State}_t + e_{t+1}$) by adding the naive count and its $1/Age$ penalty version into the state regression. The reported p-values are Newey and West (1987) (12-lag) and WCB($B = 9,999$), with the sample running from 1988:01–2026:02 (458 months).

Signal Construction	Carry Spot	Carry Total	Beta HML Spot	Beta HML Total
Naive count ≥ 2	−4.01	−1.89	−2.92	−2.99
$\text{NW}_{12}[p]$	0.096	0.457	0.295	0.308
WCB$[p]$	0.085	0.556	0.400	0.328
Naive count, excl. downcurve	−4.35	−2.76	−3.02	−2.60
$\text{NW}_{12}[p]$	0.158	0.390	0.402	0.486
WCB$[p]$	0.174	0.439	0.407	0.500
Naive count, 1/age-weighted	−14.30	−13.61	−24.10	−24.43
$\text{NW}_{12}[p]$	0.008	0.012	0.000	0.000
WCB$[p]$	0.028	0.047	0.000	0.000
IYC signal	−11.63	−9.86	−12.82	−13.58
$\text{NW}_{12}[p]$	0.003	0.013	0.001	0.000
WCB$[p]$	0.001	0.007	0.006	0.001
IYC signal, excl. downcurve	−17.35	−16.51	−17.93	−18.03
$\text{NW}_{12}[p]$	0.002	0.003	0.000	0.000
WCB$[p]$	0.002	0.004	0.001	0.001

B.3 Comparison with Other Predicting Variables and Subsample Stability Robustness Check

Tables B.8 and B.9 present the complete horse-race regression tables for Table 10, with the regression coefficients and p-values for each control variable. The change in NFCI, and the change and level of FX Volatility are the other variables that have negative and statistically significant regression coefficients, but they do not change the result for the IYC signals with and without the downcurve filter. Meanwhile, the level and the change of the VIX have positive and statistically significant regression coefficients, indicating that VIX is spiking when the portfolio starts to recover. However, they do slightly reduce the significance of the signals. Table B.10 provides the currency version of Table 1 and the currency-level horse race version of Table 9.

Table B.11 runs a leave-one-out study across the episodes, dropping one episode at a time to ensure that the result is not due to one single episode. Dropping the Great Financial

Table B.8: The carry trade portfolio horse race. The reported numbers are the regression coefficients of $\text{Return}_{t+1}^{carry} = \alpha + \gamma\,\text{State}_t + \delta' W_t + e_{t+1}$ for both the spot only and total portfolio returns. Column (i) reports the result without any controls. Column (ii) adds a control for financial conditions using NFCI; column (iii) further adds a control for the dollar factor and commodity return; column (iv) takes into account the FX Volatility; column (v) includes CLI; column (vi) adds the VIX factors; and column (vi)' applies the downcurve filter. The reported p-values are Newey and West (1987) (12-lag) and WCB($B = 9,999$). The first five columns use a sample from 1988:01–2026:02 (458 months), while the last two columns use a sample from 1990:12–2026:02 (423 months) due to data availability for the VIX.

	Spot							Total						
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vi)′	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vi)′
State	−11.63	−10.60	−10.64	−11.82	−11.83	−11.55	−15.58	−9.86	−9.04	−9.08	−10.40	−10.43	−10.66	−14.62
$\text{NW}_{12}[p]$	0.003	0.000	0.000	0.000	0.000	0.009	0.004	0.013	0.002	0.002	0.001	0.001	0.014	0.006
WCB$[p]$	0.001	0.001	0.000	0.000	0.000	0.024	0.025	0.007	0.003	0.003	0.003	0.003	0.037	0.034
Financial Conditions														
NFCI		+0.56	+0.40	+3.96	+3.92	−4.17	−5.38		+1.20	+1.06	+5.05	+4.98	−4.49	−5.59
$\text{NW}_{12}[p]$		0.870	0.907	0.401	0.404	0.430	0.267		0.746	0.778	0.303	0.310	0.371	0.226
WCB$[p]$		0.875	0.906	0.502	0.512	0.602	0.470		0.719	0.752	0.383	0.397	0.566	0.442
ΔNFCI		−46.40	−46.54	−49.01	−49.23	−69.11	−67.51		−46.50	−46.59	−49.35	−49.86	−68.16	−66.64
$\text{NW}_{12}[p]$		0.000	0.000	0.000	0.001	0.000	0.000		0.000	0.000	0.000	0.001	0.000	0.000
WCB$[p]$		0.040	0.037	0.022	0.020	0.008	0.019		0.036	0.032	0.019	0.018	0.010	0.021
Dollar and Commodity Returns														
DOL			+0.69	+0.68	+0.68	+0.46	+0.51			+0.68	+0.67	+0.67	+0.45	+0.50
$\text{NW}_{12}[p]$			0.379	0.370	0.370	0.532	0.486			0.378	0.370	0.370	0.533	0.490
WCB$[p]$			0.571	0.577	0.577	0.611	0.589			0.579	0.584	0.585	0.619	0.597
Commodity return			−0.17	−0.18	−0.18	−0.15	−0.13			−0.16	−0.18	−0.17	−0.14	−0.13
$\text{NW}_{12}[p]$			0.181	0.133	0.109	0.185	0.248			0.196	0.138	0.120	0.208	0.271
WCB$[p]$			0.367	0.330	0.266	0.389	0.424			0.396	0.350	0.289	0.413	0.446
FX Volatility														
FX volatility				−19.72	−19.75	−22.07	−20.38				−22.17	−22.24	−19.73	−18.22
$\text{NW}_{12}[p]$				0.180	0.176	0.210	0.246				0.137	0.133	0.252	0.288
WCB$[p]$				0.314	0.315	0.367	0.409				0.256	0.257	0.387	0.431
ΔFX vol.				+14.15	+14.19	+15.37	+13.25				+15.88	+15.96	+14.73	+12.77
$\text{NW}_{12}[p]$				0.314	0.312	0.311	0.382				0.262	0.258	0.336	0.403
WCB$[p]$				0.372	0.377	0.382	0.475				0.307	0.314	0.419	0.503
Leading Indicator														
ΔCLI (1m)					−0.37	−2.42	−3.22					−0.89	−3.04	−3.80
$\text{NW}_{12}[p]$					0.960	0.734	0.627					0.904	0.656	0.550
WCB$[p]$					0.990	0.924	0.887					0.976	0.900	0.856
Equity Volatility														
VIX						+0.77	+0.82						+0.74	+0.79
$\text{NW}_{12}[p]$						0.012	0.006						0.017	0.010
WCB$[p]$						0.122	0.101						0.151	0.128
ΔVIX						+0.25	+0.29						+0.26	+0.30
$\text{NW}_{12}[p]$						0.505	0.413						0.482	0.396
WCB$[p]$						0.363	0.284						0.358	0.288
$\bar{R}^2$ (%)	2.7	6.8	6.9	7.1	6.9	9.5	10.7	1.9	5.9	6.0	6.4	6.2	8.7	9.8
n	458	458	458	458	458	423	423	458	458	458	458	458	423	423

Table B.9: The beta HML portfolio horse race. The reported numbers are the regression coefficients of $\text{Return}^{\beta}_{t+1} = \alpha + \gamma\,\text{State}_t + \delta' W_t + e_{t+1}$ for both the spot only and total portfolio returns. Column (i) reports the result without any controls. Column (ii) adds a control for financial conditions using NFCI; column (iii) further adds a control for the dollar factor and commodity return; column (iv) takes into account the FX Volatility; column (v) includes CLI; column (vi) adds the VIX factors; and column (vi)' applies the downcurve filter. The reported p-values are Newey and West (1987) (12-lag) and WCB($B = 9,999$). The first five columns use a sample from 1988:01–2026:02 (458 months), while the last two columns use a sample from 1990:12–2026:02 (423 months) due to data availability for the VIX.

	Spot							Total						
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vi)′	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vi)′
State	−12.82	−12.63	−12.59	−13.38	−13.41	−15.42	−16.05	−13.58	−13.67	−13.63	−14.50	−14.52	−15.87	−16.27
$\text{NW}_{12}[p]$	0.001	0.000	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
WCB[p]	0.006	0.004	0.006	0.002	0.003	0.002	0.002	0.001	0.000	0.001	0.000	0.000	0.001	0.001
Financial Conditions														
NFCI		+2.40	+2.33	+4.73	+4.66	−7.77	−9.77		+3.15	+3.10	+5.75	+5.70	−6.76	−8.84
$\text{NW}_{12}[p]$		0.566	0.580	0.310	0.328	0.046	0.023		0.474	0.483	0.247	0.262	0.086	0.047
WCB[p]		0.413	0.452	0.348	0.366	0.111	0.069		0.283	0.323	0.280	0.300	0.169	0.112
ΔNFCI		−37.20	−38.99	−42.95	−43.48	−52.98	−51.86		−35.92	−37.64	−41.57	−41.95	−51.52	−50.43
$\text{NW}_{12}[p]$		0.024	0.028	0.025	0.028	0.018	0.021		0.032	0.038	0.034	0.038	0.024	0.028
WCB[p]		0.215	0.210	0.166	0.135	0.178	0.213		0.235	0.232	0.185	0.159	0.193	0.232
Dollar and Commodity Returns														
DOL			−0.63	−0.63	−0.63	−0.57	−0.52			−0.69	−0.68	−0.68	−0.59	−0.53
$\text{NW}_{12}[p]$			0.357	0.363	0.362	0.412	0.469			0.317	0.322	0.321	0.405	0.462
WCB[p]			0.672	0.660	0.660	0.692	0.734			0.613	0.599	0.599	0.684	0.726
Commodity return			+0.05	+0.05	+0.05	+0.00	+0.03			+0.07	+0.06	+0.07	+0.02	+0.05
$\text{NW}_{12}[p]$			0.766	0.772	0.721	0.979	0.840			0.694	0.704	0.655	0.881	0.751
WCB[p]			0.816	0.825	0.788	0.981	0.854			0.765	0.775	0.741	0.895	0.778
FX Volatility														
FX volatility				−13.32	−13.38	−6.11	−3.00				−14.71	−14.76	−6.00	−2.75
$\text{NW}_{12}[p]$				0.312	0.302	0.627	0.826				0.293	0.285	0.634	0.843
WCB[p]				0.337	0.326	0.828	0.871				0.341	0.331	0.797	0.875
ΔFX vol.				+16.23	+16.31	+12.96	+10.20				+16.64	+16.69	+12.33	+9.50
$\text{NW}_{12}[p]$				0.171	0.168	0.343	0.467				0.175	0.173	0.379	0.511
WCB[p]				0.094	0.085	0.259	0.382				0.093	0.082	0.281	0.427
Leading Indicator														
ΔCLI (1m)					−0.93	−0.65	−1.24					−0.67	−0.80	−1.38
$\text{NW}_{12}[p]$					0.900	0.925	0.852					0.928	0.906	0.834
WCB[p]					0.938	0.942	0.889					0.957	0.933	0.881
Equity Volatility														
VIX						+0.89	+0.94						+0.85	+0.91
$\text{NW}_{12}[p]$						0.001	0.000						0.001	0.001
WCB[p]						0.065	0.047						0.069	0.051
ΔVIX						+0.02	+0.06						+0.04	+0.08
$\text{NW}_{12}[p]$						0.966	0.899						0.932	0.864
WCB[p]						0.965	0.895						0.925	0.857
$\bar{R}^2$ (%)	2.7	4.7	4.4	4.4	4.1	7.2	6.8	3.0	4.9	4.7	4.7	4.5	6.9	6.5
n	458	458	458	458	458	423	423	458	458	458	458	458	423	423

Table B.10: Robustness check on country's own inversion and other yield curve variables. This table presents a currency level and pooled regression of: $\Delta s_{i,t+1} = \alpha + \gamma\,\text{State}_t + e_{i,t+1}$. *ON* column shows how many months that each of the currency's yield curve are inverted; *Own Only* column shows how many of those inversions are active outside the IYC signal; the next columns provide the regression coefficient of the country's own inversion, the ones without IYC, the IYC signal, and the IYC signal controlling the change in slope, relative slope against the United States following the Nelson-Siegel slope factor from Chen and Tsang (2013). The reported p-values at the currency level are the Newey and West (1987) (12-lag) and WCB ($B = 9,999$), while the pooled version replaces Newey and West (1987) with **?** (12-lag) to take into account the currency fixed effects. The sample runs from 1998:01–2026:02 (458 months).

Currency	ON	Own Only	Own	Own, No IYC	IYC signal	IYC signal, Ctrl.
AUD	44	3	−5.14	—	−11.53	−13.08
$NW_{12}[p]$			0.267	—	0.056	0.084
WCB$[p]$			0.305	—	0.056	0.156
CAD	41	10	−1.81	−1.98	−3.68	−4.02
$NW_{12}[p]$			0.621	0.294	0.260	0.353
WCB$[p]$			0.650	0.360	0.342	0.502
CHF	12	0	−11.91	—	+1.45	+1.93
$NW_{12}[p]$			0.195	—	0.748	0.733
WCB$[p]$			0.416	—	0.738	0.736
EUR	28	0	+4.65	—	+1.45	+1.31
$NW_{12}[p]$			0.562	—	0.775	0.836
WCB$[p]$			0.596	—	0.778	0.843
GBP	68	21	−6.15	−8.13	−2.79	−3.75
$NW_{12}[p]$			0.219	0.109	0.554	0.523
WCB$[p]$			0.228	0.142	0.593	0.597
JPY	22	2	−1.74	—	−0.39	+2.42
$NW_{12}[p]$			0.694	—	0.920	0.535
WCB$[p]$			0.797	—	0.926	0.568
NOK	70	27	−4.58	+18.09	−4.77	−8.94
$NW_{12}[p]$			0.450	0.001	0.395	0.218
WCB$[p]$			0.549	0.013	0.445	0.293
NZD	54	12	−1.89	+4.00	−13.71	−12.59
$NW_{12}[p]$			0.738	0.689	0.006	0.052
WCB$[p]$			0.785	0.618	0.008	0.112
SEK	24	0	−10.75	—	−3.15	−5.84
$NW_{12}[p]$			0.115	—	0.545	0.373
WCB$[p]$			0.175	—	0.577	0.469
Pooled G10			−4.00	+2.90	−4.13	−4.82
WCB$[p]$			0.159	0.503	0.335	0.411
DK$[p]$			0.136	0.479	0.284	0.323

Crisis (GFC) and COVID-19 episodes gives the biggest impact for the results, but it is still very robust.

Table B.11: Leave one episode out robustness check. The signal active in 14 different episodes when splitting them by the non-signal months. So, this table provides a study that removes each one of those episode to ensure that the result is not a "one episode result". After removing the observations in that one episode, I re-estimate the IYC regression: $\text{Return}_{t+1} = \alpha + \gamma\,\text{IYC}_t + e_{t+1}$. The reported p-value in the bracket is WCB ($B = 9,999$), with the sample running from 1988:01-2026:02 (458 months).

Episode Dropped	IYC Months	Carry Spot	Carry Total	Beta HML Spot	Beta HML Total
None (baseline)	—	−11.63 [0.001]	−9.86 [0.007]	−12.82 [0.006]	−13.58 [0.001]
1988:08–1990:09	26	−13.39 [0.003]	−12.67 [0.008]	−17.55 [0.002]	−17.91 [0.001]
1990:12–1991:06	7	−12.30 [0.002]	−10.42 [0.008]	−11.13 [0.012]	−11.74 [0.002]
1991:08–1991:08	1	−11.67 [0.002]	−9.87 [0.008]	−12.79 [0.009]	−13.54 [0.001]
1995:01–1995:02	2	−11.18 [0.002]	−9.39 [0.010]	−12.69 [0.010]	−13.47 [0.002]
1997:08–1998:05	10	−11.90 [0.003]	−10.09 [0.012]	−11.92 [0.011]	−12.83 [0.001]
1998:08–1998:10	3	−10.37 [0.003]	−8.59 [0.016]	−12.96 [0.008]	−13.66 [0.001]
2000:04–2000:10	7	−11.33 [0.003]	−9.46 [0.014]	−12.60 [0.012]	−13.50 [0.001]
2000:12–2001:01	2	−12.33 [0.001]	−10.53 [0.005]	−13.41 [0.005]	−14.21 [0.000]
2005:03–2005:03	1	−11.83 [0.001]	−10.05 [0.008]	−12.66 [0.006]	−13.40 [0.001]
2006:01–2006:12	12	−13.06 [0.001]	−11.19 [0.004]	−13.48 [0.011]	−14.09 [0.001]
2007:08–2007:09	2	−11.56 [0.001]	−9.77 [0.009]	−13.20 [0.006]	−14.01 [0.001]
2008:06–2008:10	5	−8.99 [0.004]	−7.23 [0.023]	−10.98 [0.013]	−11.86 [0.002]
2020:02–2020:03	2	−10.65 [0.002]	−8.80 [0.013]	−11.82 [0.008]	−12.59 [0.000]
2022:08–2023:08	13	−13.06 [0.001]	−10.95 [0.008]	−13.42 [0.012]	−14.38 [0.001]

Table B.12 splits the sample at the end of 2004 to check whether the result is robust before and after the GFC. Interestingly, the result for the carry trade is statistically stronger prior to the GFC, but the one that includes the GFC is better at capturing the tail events.

B.4 The Persistence of the Signal

The IYC signal is a persistent signal, where the 93 signal months actually come only around 5-14 episodes (depending on what is a definition of an episode). Table B.13 provides a robustness check on different definition of an episode, the baseline split is using non-signal months as a separator (14 episodes), meanwhile an alternative definitions are if the cluster of the signals are within 6 (10 episodes) and 12 months (7 episodes) with each other, and lastly

Table B.12: Split sample robustness check. This table presents a robustness check by splitting the sample into two. The middle of the sample occurs around the GFC, so instead I split it at the end of 2004 to make sure that one sample contains GFC and the other one does not. For each sample, the following regression coefficients are provided: $\text{Return}_{t+1} = \alpha + \gamma \text{State}_{t+1} + e_{t+1}$, where the State follows the construction from (12). The bottom panel provides the number of months that each of the sample have, the number of IYC signal and downcurve months, the number of episodes by the baseline definition, and how many of the three worst carry trade months are captures in the sample. The reported p-values are Newey and West (1987) (12-lag) and WCB ($B = 9,999$), with sample running from 1988:01–2026:02.

	1988:01–2004:12	2005:01–2026:02
Carry, spot	−12.21	−12.64
$\text{NW}_{12}[p]$	0.002	0.143
WCB$[p]$	0.017	0.055
non-downcurve	−16.99	−17.94
$\text{NW}_{12}[p]$	0.002	0.079
WCB$[p]$	0.030	0.028
downcurve	−3.77	+9.62
$\text{NW}_{12}[p]$	0.237	0.005
WCB$[p]$	0.067	0.109
Carry, total	−10.94	−11.27
$\text{NW}_{12}[p]$	0.008	0.185
WCB$[p]$	0.025	0.230
non-downcurve	−17.10	−16.47
$\text{NW}_{12}[p]$	0.001	0.105
WCB$[p]$	0.031	0.116
downcurve	−1.61	+10.33
$\text{NW}_{12}[p]$	0.621	0.003
WCB$[p]$	0.210	0.103
Beta HML, spot	−13.56	−14.57
$\text{NW}_{12}[p]$	0.008	0.020
WCB$[p]$	0.017	0.008
non-downcurve	−16.05	−20.21
$\text{NW}_{12}[p]$	0.000	0.007
WCB$[p]$	0.040	0.001
downcurve	−6.85	+9.38
$\text{NW}_{12}[p]$	0.356	0.381
WCB$[p]$	0.658	0.290
Beta HML, total	−14.31	−15.02
$\text{NW}_{12}[p]$	0.005	0.013
WCB$[p]$	0.009	0.007
non-downcurve	−15.40	−20.94
$\text{NW}_{12}[p]$	0.000	0.003
WCB$[p]$	0.042	0.001
downcurve	−8.68	+10.08
$\text{NW}_{12}[p]$	0.245	0.352
WCB$[p]$	0.378	0.283
n	204	254
Signal months (downcurve)	58 (30)	35 (7)
Episodes	8	6
Three worst carry months inside	1 of 3	3 of 3

using the trough to trough of NBER business cycle (5 episodes). This distinction matters in doing the WCB test since it is a function of the number of clusters or episodes. The strictest test which is a leave-one-out regression for the NBER business cycle shows that the results are robust at 5% significant level except when we leave out 1991-2001 cycle, where the carry trade total return lose the significance. This is interesting per se, because people would think the biggest driver of the result would be the GFC or even COVID-19, which is clearly not the case here.

Table B.13: Episodes robustness check. This table shows the results for different definition of an episodes. *Panel A*: The baseline result uses non-signal months as the separator; alternative definition includes merging the signals that are within 6 and 12 months with each other, and to use the NBER business cycle trough-to-trough. *Panel B*: leave-one-out regression for the NBER business cycle episode definition. *Panel C*: randomization inference for the results.

Panel A: Definition of an Episode	# of Episodes	Carry spot	Carry total	Beta spot	Beta total
γ (%/yr), identical for all		−11.63	−9.86	−12.82	−13.58
WCB[p] *on* γ					
Signal episodes (baseline)	14	0.001	0.007	0.006	0.001
Merging episodes within 6 months	10	0.001	0.010	0.001	0.001
Merging episodes within 12 months	7	0.002	0.011	0.001	0.001
NBER business cycles	5	0.023	0.017	0.024	0.024
Panel B: Business Cycle Leave-One-Out, γ *(%/yr) with WCB*[p] *in brackets*					
	n (IYC)	Carry spot	Carry total	Beta spot	Beta total
1988:01–1991:03	39 (30)	−14.23 [0.002]	−13.38 [0.006]	−15.81 [0.001]	−15.87 [0.001]
1991:03–2001:11	128 (28)	−9.85 [0.024]	−7.32 [0.109]	−10.05 [0.065]	−11.34 [0.013]
2001:11–2009:06	91 (20)	−9.33 [0.007]	−7.24 [0.036]	−11.94 [0.022]	−12.74 [0.005]
2009:06–2020:04	130 (2)	−12.31 [0.002]	−11.05 [0.003]	−13.76 [0.002]	−14.43 [0.000]
2020:04–2026:02	70 (13)	−13.15 [0.001]	−11.28 [0.010]	−13.38 [0.014]	−14.32 [0.002]
Panel C: Randomization Inference, p *on the headline* γ					
Placement	Draws	Carry spot	Carry total	Beta spot	Beta total
Episodes relocated, p on γ	9,999	0.001	0.003	0.000	0.000
Episodes relocated, p on t	9,999	0.003	0.015	0.001	0.000
Circular shifts, p on t	457	0.015	0.031	0.004	0.002

B.5 Beta Definitions

Table B.14 conducts a robustness check for the beta HML portfolio construction. It provides an alternative measure for calculating the currency beta, using a 24-month rolling window

instead of 90 days to see if the result holds by changing the data frequency, especially since the study is conducted using monthly data. Additionally, I change the proxy for the equity market by using MSCI World instead of S&P 500 to capture the global equity market instead of the United States. The results hold qualitatively across the board, with only the 90-day rolling window for MSCI World losing some of the significance.

Table B.14: Reconstructing the beta HML portfolio using an alternative currency beta calculation. The baseline definition for currency beta in the main paper uses a 90-day rolling window (4), while this table provides an alternative calculation using a 24-month rolling window and the MSCI World. The reported coefficients are the results from regressions $X^{\beta}_{t+1} = \alpha + \gamma\,\text{State}_t + e_{t+1}$ for the spot only return, and $r^{\beta}_{t+1} = \alpha + \gamma\,\text{State}_t + e_{t+1}$ for the total portfolio return including IRD. The reported p-values are Newey and West (1987) (12-lag) and WCB ($B = 9,999$), with the sample running from 1988:01–2026:02 (458 months).

Beta Definition	n	Signal, Spot	Signal, Total	Non-downcurve, Spot	Non-downcurve, Total
Daily 90-day rolling, S&P 500 (baseline)	458	−12.82	−13.58	−17.93	−18.03
$NW_{12}[p]$		0.001	0.000	0.000	0.000
WCB[p]		0.006	0.001	0.001	0.001
Monthly 24-month rolling, S&P 500	458	−9.79	−9.62	−17.62	−18.38
$NW_{12}[p]$		0.022	0.025	0.000	0.000
WCB[p]		0.031	0.022	0.002	0.000
Daily 90-day rolling, MSCI World	458	−7.73	−10.24	−10.19	−10.60
$NW_{12}[p]$		0.061	0.009	0.078	0.062
WCB[p]		0.103	0.004	0.086	0.062
Monthly 24-month rolling, MSCI World	458	−10.11	−10.45	−15.13	−15.69
$NW_{12}[p]$		0.007	0.004	0.003	0.001
WCB[p]		0.013	0.013	0.014	0.001

B.6 Portfolio Composition by State

Following the result from Section 3.2, the natural question is whether the portfolio composition during the signal months follows the sample average ranking in Table 4. So, Table B.15 shows the 3×3 carry trade portfolio composition across different signals. It shows that the top-two and bottom-two currencies are consistent across the states, with NZD and AUD the two currencies with significant currency level results for the signal are the two most long currencies, while safe haven currencies JPY and CHF occupy all the short leg. A couple of

notable changes during the downcurve periods are that GBP becomes a prominent investment currency in place of NOK, and EUR becomes a high-frequency funding currency in place of SEK.

Table B.15: Carry trade portfolio composition by signal state. The reported column are the full sample composition, during the signal states (IYC signal, IYC signal excluding the downcurve months, and IYC signal with the downcurve months). The currency is sorted by IRD, with the sample is from 1988:01-2026:02 (458 months).

	Long Leg (% of Months)				Short Leg (% of Months)			
	Full	IYC signal	Non-downcurve	Downcurve	Full	IYC signal	Non-downcurve	Downcurve
NZD	85	92	95	89	2	0	0	0
AUD	71	71	71	70	4	0	0	0
NOK	62	30	48	3	3	4	7	0
GBP	43	74	59	97	4	0	0	0
CAD	21	19	21	16	23	12	18	3
SEK	15	13	5	24	33	35	48	16
EUR	4	0	0	0	51	48	27	81
JPY	0	0	0	0	83	100	100	100
CHF	0	0	0	0	96	100	100	100

B.7 Panel Interactions outside the G10

Table B.16 repeats the study of (13) and (14) on the seven EM currencies. The results from G10 only do not hold here, as there is more noise in EM beta and IRD (the coefficient is positive in EM only instead of negative).

B.8 The EM Curves

Table B.17 rebuilds the IYC signal using only the seven EM yield curves, then applies it to EM+G10 carry trade portfolio. It presents stylized facts about the spot leg, income leg, and the total return across different signals. There are only seven downcurve months (in

Table B.16: Signal interaction with EM. This table repeats the study from Table 5 for EM currencies and EM+G10 currencies using (13) & (14). Column (i) provides the pooled result of the IYC signal across the currencies without any interaction. Column (ii) & (iv) provide the results for the interaction with real-time beta and IRD. Column (iii) & (v) provide the results for the interaction with sample average beta and IRD. The reported p-values are Driscoll and Kraay (1998) (12-lag) and WCB($B = 9,999$), with the sample running from 1996:02–2026:02 (361 months) due to EM data availability.

	(i)	(ii)	(iii)	(iv)	(v)
Panel A: Seven EM Currencies					
IYC_t	−13.87	−14.62	−18.94	−12.08	−16.09
$\text{DK}_{12}[p]$	0.082	0.137	0.196	0.154	0.034
$\text{WCB}[p]$	0.175	0.182	0.471	0.238	0.018
$\hat{\beta}_{i,t}$		−7.86			
$\text{DK}_{12}[p]$		0.475			
$\text{WCB}[p]$		0.532			
$\text{IYC}_t \times \hat{\beta}_{i,t}$		+3.32			
$\text{DK}_{12}[p]$		0.886			
$\text{WCB}[p]$		0.830			
$\text{IYC}_t \times \bar{\beta}_i$			+27.13		
$\text{DK}_{12}[p]$			0.590		
$\text{WCB}[p]$			0.887		
$\text{IRD}_{i,t}$				−0.05	
$\text{DK}_{12}[p]$				0.865	
$\text{WCB}[p]$				0.849	
$\text{IYC}_t \times \text{IRD}_{i,t}$				−0.24	
$\text{DK}_{12}[p]$				0.827	
$\text{WCB}[p]$				0.696	
$\text{IYC}_t \times \overline{\text{IRD}}_i$					+0.37
$\text{DK}_{12}[p]$					0.560
$\text{WCB}[p]$					0.549
n	2527	2527	2527	2527	2527
Panel B: Sixteen EM+G10 Currencies					
IYC_t	−9.55	−9.73	−5.55	−7.53	−6.93
$\text{DK}_{12}[p]$	0.112	0.129	0.304	0.137	0.170
$\text{WCB}[p]$	0.189	0.208	0.453	0.179	0.241
$\hat{\beta}_{i,t}$		+1.14			
$\text{DK}_{12}[p]$		0.868			
$\text{WCB}[p]$		0.851			
$\text{IYC}_t \times \hat{\beta}_{i,t}$		+2.05			
$\text{DK}_{12}[p]$		0.915			
$\text{WCB}[p]$		0.920			
$\text{IYC}_t \times \bar{\beta}_i$			−30.83		
$\text{DK}_{12}[p]$			0.123		
$\text{WCB}[p]$			0.187		
$\text{IRD}_{i,t}$				+0.16	
$\text{DK}_{12}[p]$				0.555	
$\text{WCB}[p]$				0.665	
$\text{IYC}_t \times \text{IRD}_{i,t}$				−0.77	
$\text{DK}_{12}[p]$				0.387	
$\text{WCB}[p]$				0.502	
$\text{IYC}_t \times \overline{\text{IRD}}_i$					−1.01
$\text{DK}_{12}[p]$					0.127
$\text{WCB}[p]$					0.256
n	5776	5776	5776	5776	5776

The sixteen-currency beta HML portfolio, on-signal spot effect (%/yr)					
re-sorted monthly on the live beta	−24.29	$\text{NW}_{12}[p]$ 0.023	$\text{WCB}[p]$ 0.019		
frozen sample-average ranking	−12.34	$\text{NW}_{12}[p]$ 0.050	$\text{WCB}[p]$ 0.071		

Table B.17: EM yield curves applied to G10+EM portfolio. This table repeats the study of the IYC signal from Table 3 ($\text{Return}_{t+1} = \alpha + \gamma\,\text{State}_t + e_{t+1}$) by constructing the signals using seven EM yield curves (State_t^{EM}) for the portfolio return constructed using sixteen G10+EM currencies ($\text{Return}_{t+1}^{G10+EM}$). The reported p-values are Newey and West (1987) (12-lag) and WCB($B = 9,999$), with the sample running from 2003:07–2026:02 (272 months) due to EM data availability.

State	n	Spot	IRD	Total
EM signal on, EM downcurve months	7	+17.41	+10.31	+27.71
EM signal on, EM non-downcurve months	71	−7.18	+9.85	+2.67
EM signal off	194	+0.84	+8.26	+9.10
State regressions on total returns, each state against every other month				
EM signal (γ)				−4.18
$\text{NW}_{12}[p]$				0.431
WCB[p]				0.553
EM signal, non-downcurve (γ)				−7.08
$\text{NW}_{12}[p]$				0.160
WCB[p]				0.283
EM signal, downcurve (γ)				+20.34
$\text{NW}_{12}[p]$				0.009
WCB[p]				0.154

2023–2024) in the sample, since it starts in July 2003 due to EM data availability, but it shows that during those periods, the performance of the spot leg, and thus the carry trade is the best by a huge margin. The bottom panel provides the regression coefficients for each of the signals from (12). The results hold qualitatively from the G10 only portfolio, but with weaker power, mostly due to the high IRD from EM currencies as part of the analogy for the *overwrite*.

B.9 The Common Term Premium

An alternative explanation for the inversion, it is driven by a common global term premium, using Adrian et al. (2013) US slope premium as a proxy. Using the measures as a control variable does not change the result, but Table B.18 reports an alternative study that strips each country's slope of its fitted exposure to the US slope and rebuilds the signal based on those stripped slopes. There are a total of 1,122 months of individual countries' signal

Table B.18: The signal is rebuilt by stripping the term premium from the slopes. Each country's 10Y–2Y slope is stripped of its fitted exposure to the US slope term premium using the Adrian et al. (2013) method. Each of the cells is a regression of the portfolio return on the IYC signal (unstripped, stripped, and split between downcurve and non-downcurve regimes) $r_{t+1} = \alpha + \gamma State_t + e_{t+1}$. The reported result is annualized (%/yr); the reported p-values are Newey-West (NW_{12}) and wild cluster bootstrap (WCB); sample runs from 1988:01-2026:02.

State	Carry, Spot	Carry, Total	Beta HML, Spot	Beta HML, Total
Baseline signal	−11.63	−9.86	−12.82	−13.58
$NW_{12}[p]$	0.003	0.013	0.001	0.000
WCB[p]	0.001	0.007	0.006	0.001
Stripped signal	−4.10	−2.69	−8.00	−8.39
$NW_{12}[p]$	0.123	0.309	0.001	0.001
WCB[p]	0.127	0.370	0.034	0.020
non-downcurve months	−5.44	−4.71	−8.54	−8.52
$NW_{12}[p]$	0.111	0.164	0.002	0.002
WCB[p]	0.126	0.203	0.009	0.011
downcurve months	+0.27	+2.46	−3.65	−4.69
$NW_{12}[p]$	0.923	0.301	0.475	0.371
WCB[p]	0.902	0.101	0.790	0.826
downcurve − other	+4.66	+6.08	+3.59	+2.62
$NW_{12}[p]$	0.277	0.145	0.499	0.638
WCB[p]	0.221	0.117	0.552	0.694

months; when we rebuild using the stripped term premium, it leaves the original 937 months, while adding 110 months outside the original construction. Meanwhile, the rebuilt IYC signal has 124 months compared to the original 93 months. The qualitative results hold across carry trade and beta HML portfolios, just becoming less significant.

B.10 Local Projections After a Signal Month

The IYC signal is persistent, so doing a local projection on it leads to an overlapping data issue. A cleaner way to deal with this is to do the local projection on the onset of the signal or control for the lags of the signal. Figure B.2 does all of these studies and shows that the CLI's decline is concentrated in the first 3 to 6 months after the IYC signal.

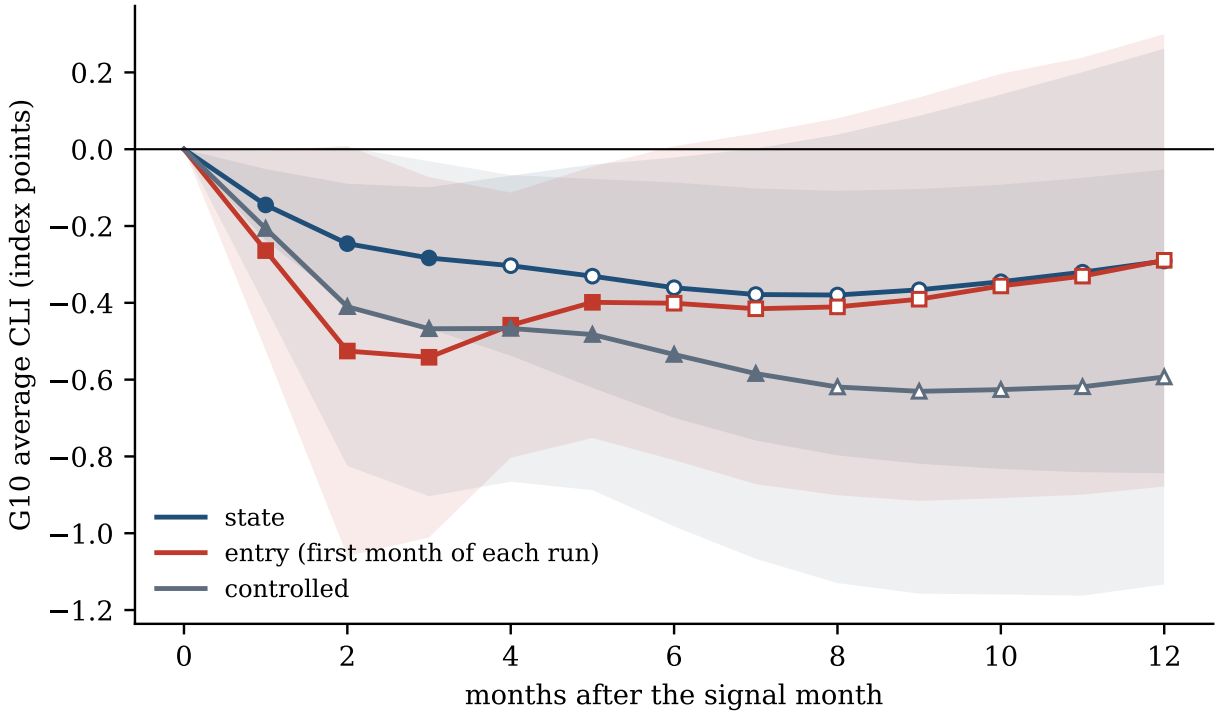


Figure B.2: Local projections of the G10's CLI average after the signal months. The circles mark the IYC signal, the squares mark the onset of the IYC signal, and the triangles mark the IYC signal controlling for its own lags; the reported bands are the 90% Newey–West confidence intervals.